16. $\displaystyle\int_{-\infty}^{\infty} x^2 \mathbf{e}^{x^3}\, dx$

17. $\displaystyle\int_{-\infty}^{\infty} \frac{y}{\left(y^2+1\right)^3}\, dy$

18. $\displaystyle\int_{0}^{3} \frac{w^3}{\sqrt{9-w^2}}\, dw$

19. $\displaystyle\int_{-3}^{1} \frac{1}{w^2+2w}\, dw$

20. $\displaystyle\int_{0}^{\infty} \frac{\mathbf{e}^{\frac{1}{x}}}{x^2}\, dx$

21. $\displaystyle\int_{0}^{\infty} \frac{1}{z[\ln(z)]^2}\, dz$

22. $\displaystyle\int_{0}^{\infty} \frac{1}{w-1}\, dw$

7.9 Comparison Test for Improper Integrals

Use the Comparison Test to determine if the following integrals converge or diverge.

1. $\displaystyle\int_4^\infty \frac{1}{\sqrt[5]{z-2}}\,dz$

2. $\displaystyle\int_0^\infty \frac{w}{\sqrt{w^6+2}}\,dw$

3. $\displaystyle\int_2^\infty \frac{1}{(2w+3)^4}\,dw$

4. $\displaystyle\int_{12}^\infty \frac{y^2-4y+2}{y-7}\,dy$

5. $\displaystyle\int_2^\infty \frac{1}{\ln(x)}\,dx$
   Hint : Sketch the graph of $y=x$ and $y=\ln(x)$ on the same axis system.

6. $\displaystyle\int_2^\infty \frac{\sqrt{z}-4\sin^2(z)}{z^3}\,dz$

7. $\displaystyle\int_{20}^\infty \frac{\sqrt[3]{2x}+\sin^2(x)}{\sqrt{x}-\cos^2(x)}\,dx$

8. $\displaystyle\int_0^\infty \frac{z\mathbf{e}^{-z}}{z^3+1}\,dz$

7.10 Approximating Definite Integrals

For each of the following integrals use the given value of n to approximate the value of the definite integral using

(a) the Midpoint Rule,

(b) the Trapezoid Rule, and

(c) Simpson's Rule.

Use at least 6 decimal places of accuracy for your work.

1. $\int_{-2}^{4} \sin\left(x^2+2\right)\, dx$ using $n=6$
2. $\int_{0}^{4} \sqrt[3]{x^4+6}\, dx$ using $n=6$
3. $\int_{1}^{5} \mathbf{e}^{\cos(x)}\, dx$ using $n=8$
4. $\int_{3}^{5} \frac{1}{1-\ln(x)}\, dx$ using $n=6$
5. $\int_{-3}^{1} \sin(x)\cos\left(x^2\right)\, dx$ using $n=8$

8 More Applications of Integrals

It is now time to take a look at some more applications of integrals. As noted the last time we looked at applications of integrals many, although, not all of these new applications in this chapter have a fairly high chance of needing some of the integration techniques from the last chapter.

The first application, Arc Length can be kept to only u-substitutions at the worst, but most of those problems tend to be very simple. Once we start moving into more complicated problems arc length problems tend to involve trig substitutions.

The next application, Surface Area tends to be u-substitutions but the notation used here is also used in the Arc Length section and so the surface area section is also here because of the shared notation.

Center of Mass and Probability are applications that will, in almost every case, involve integration by parts. In addition, the Probability section has the potential for improper integrals to show up.

The other application we'll be looking at in this chapter, Hydrostatic Pressure and Force, will typically involve fairly simple integrals that could have been done in the earlier chapter. The reason the topic is here is because we have to derive up the integral using the definition of the definite integral in every problem. In addition, more complicated problems could lead to much more complicated integrals. The integrals in this section are kept simple mostly to keep the derivation work simpler.

If you are looking for some practice problems (with solutions available) please check out the Practice Problems. There you will find a set of problems that should give you quite a bit practice.

8.1 Arc Length

1. Set up, but do not evaluate, an integral for the length of $y = 14 - 9x$, $-22 \le y \le 31$ using,

   (a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$

   (b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

2. Set up, but do not evaluate, an integral for the length of $x = \mathbf{e}^{2y}$, $-1 \le y \le 0$ using,

   (a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$

   (b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

3. Set up, but do not evaluate, an integral for the length of $y = \tan(2x)$, $0 \le x \le \frac{\pi}{3}$ using,

   (a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$

   (b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

4. Set up, but do not evaluate, an integral for the length of $\frac{x^2}{16} + 9y^2 = 1$.

5. For $x = 6y + 1$, $-2 \le y \le 8$

   (a) Use an integral to find the length of the curve.

   (b) Verify your answer from part (a) geometrically.

6. Determine the length of $y = \frac{4}{3}x + 2$, $0 \le x \le 9$.

7. Determine the length of $y = (8x + 3)^{\frac{3}{2}}$, $11^{\frac{3}{2}} \le y \le 27^{\frac{3}{2}}$.

8. Determine the length of $x = (10 - 2y)^{\frac{3}{2}}$, $-1 \le y \le 2$.

9. Determine the length of $x = 2 + (y - 1)^2$, $2 \le y \le 5$.

10. Determine the length of $y = (3x + 2)^2$, $1 \le x \le 4$.

8.2 Surface Area

1. Set up, but do not evaluate, an integral for the surface area of the object obtained by rotating $y = 7x + 2$, $-5 \le y \le 0$ about the x-axis using,

   (a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$

   (b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

2. Set up, but do not evaluate, an integral for the surface area of the object obtained by rotating $y = 1 + 2x^5$, $0 \le x \le 1$ about the x-axis using,

   (a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$

   (b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

3. Set up, but do not evaluate, an integral for the surface area of the object obtained by rotating $x = \mathbf{e}^{2y}$, $-1 \le y \le 0$ about the y-axis using,

   (a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$

   (b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

4. Set up, but do not evaluate, an integral for the surface area of the object obtained by rotating $y = \cos\left(\frac{1}{2}x\right)$, $0 \le x \le \pi$ about

   (a) the x-axis

   (b) the y-axis

5. Set up, but do not evaluate, an integral for the surface area of the object obtained by rotating $x = \sqrt{3 + 7y}$, $0 \le y \le 1$ about

   (a) the x-axis

   (b) the y-axis

6. Find the surface area of the object obtained by rotating $y = \frac{1}{4}\sqrt{6x + 2}$, $\frac{\sqrt{2}}{2} \le y \le \frac{\sqrt{5}}{2}$ about the x-axis.

7. Find the surface area of the object obtained by rotating $y = 4 - x$, $1 \leq x \leq 6$ about the y-axis.

8. Find the surface area of the object obtained by rotating $x = 2y + 5$, $-1 \leq x \leq 2$ about the y-axis.

9. Find the surface area of the object obtained by rotating $x = 1 - y^2$, $0 \leq y \leq 3$ about the x-axis.

10. Find the surface area of the object obtained by rotating $x = \mathbf{e}^{2y}$, $-1 \leq y \leq 0$ about the y-axis.

11. Find for the surface area of the object obtained by rotating $y = \cos\left(\frac{1}{2}x\right)$, $0 \leq x \leq \pi$ about the x-axis.

8.3 Center Of Mass

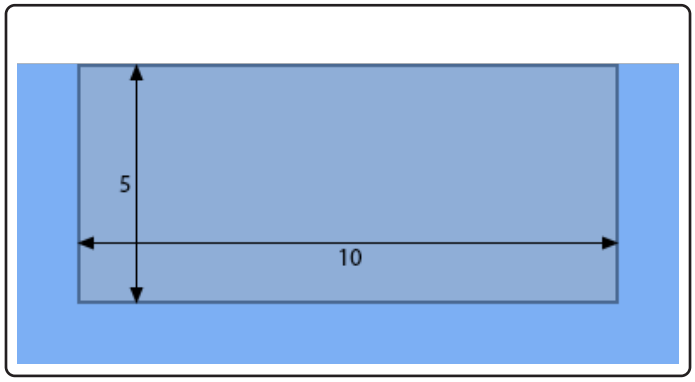
Find the center of mass for each of the following regions.

1. The region bounded by $y = x^3$, $x = -2$ and the x-axis.
2. The triangle with vertices $(-2, -2)$, $(4, -2)$ and $(4, 4)$.
3. The region bounded by $y = (x - 2)^2$ and $y = 4$.
4. The region bounded by $y = \cos(x)$ and the x-axis between $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.
5. The region bounded by $y = x^2$ and $y = 6 - x$.
6. The region bounded by $y = \mathbf{e}^{2x}$ and the x-axis between $-1 \leq x \leq 1$.
7. The region bounded by $y = \mathbf{e}^{2x}$ and $y = -\cos(\pi x)$ between $-\frac{1}{2} \leq x \leq \frac{1}{2}$.

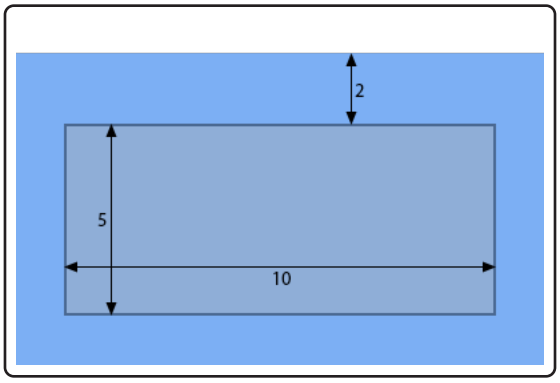
8.4 Hydrostatic Pressure and Force

Find the hydrostatic force on the following plates submerged in water as shown in each image. In each case consider the top of the blue “box” to be the surface of the water in which the plate is submerged. Note as well that the dimensions in many of the images will not be perfectly to scale in order to better fit the plate in the image. The lengths given in each image are in meters.

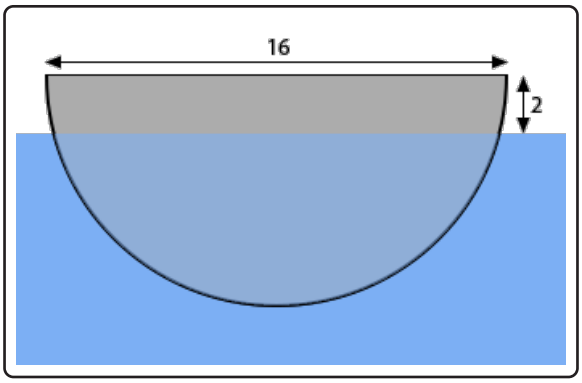
1.



2.



3.



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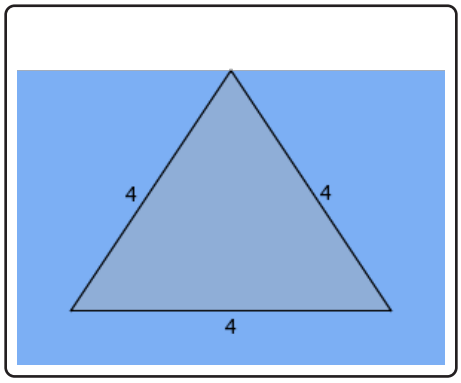
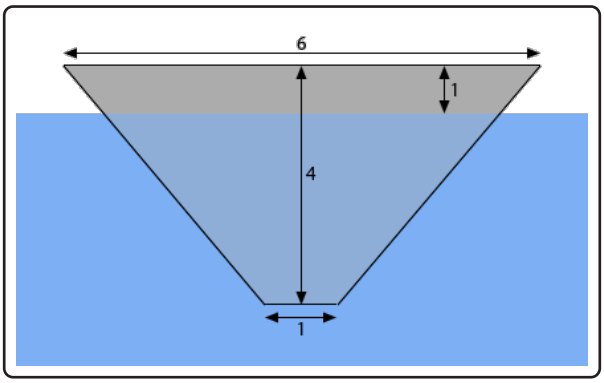
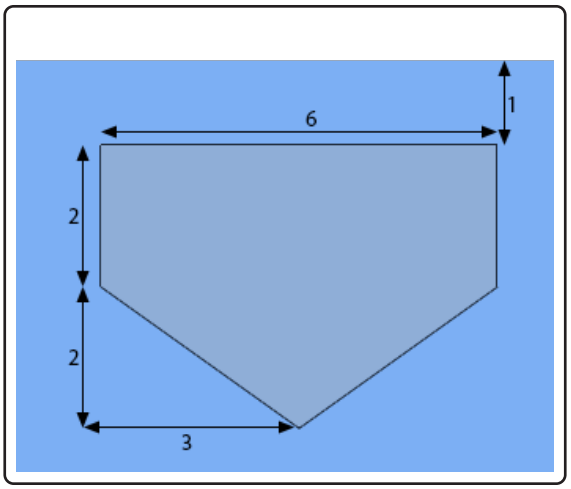


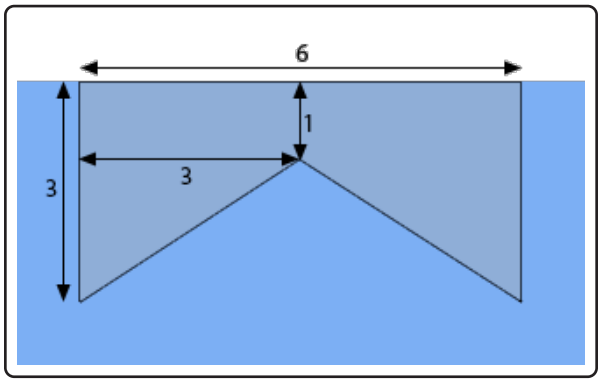
5.



6.


6
1
2
2
3

7.



8.5 Probability

1. Let,

$$f(x) = \begin{cases} \frac{3}{4}\left(2x - x^2\right) & \text{if } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

   (a) Show that $f(x)$ is a probability density function.

   (b) Find $P(X \leq 0.25)$.

   (c) Find $P(X \geq 1.4)$.

   (d) Find $P(0.1 \leq X \leq 1.2)$.

   (e) Determine the mean value of X.

2. Let,

$$f(x) = \begin{cases} \frac{4}{\ln(3)\left(4x + x^2\right)} & \text{if } 1 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

   (a) Show that $f(x)$ is a probability density function.

   (b) Find $P(X \leq 1)$.

   (c) Find $P(X \geq 5)$.

   (d) Find $P(1 \leq X \leq 5)$.

   (e) Determine the mean value of X.

3. Let,

$$f(x) = \begin{cases} \frac{1}{10}\left(1 + \sin\left(\pi x - \frac{\pi}{2}\right)\right) & \text{if } 0 \leq x \leq 10 \\ 0 & \text{otherwise} \end{cases}$$

   (a) Show that $f(x)$ is a probability density function.

   (b) Find $P(X \leq 3)$.

   (c) Find $P(X \geq 5)$.

   (d) Find $P(2.5 \leq X \leq 7)$.

   (e) Determine the mean value of X.

4. The probability density function of the life span of a battery is given by the function below, where t is in years.

$$f(t) = \begin{cases} 1.25\mathbf{e}^{-1.25t} & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

   (a) Verify that $f(t)$ is a probability density function.

   (b) What is the probability that a battery will have a life span less than 10 months?

   (c) What is the probability that a battery will have a life span more than 2 years?

   (d) What is the probability that a battery will have a life span between 1.5 and 4 years?

   (e) Determine the mean value of the life span of the batteries.

5. The probability density function of the successful outcome from some experiment is given by the function below, where t is in minutes.

$$f(t) = \begin{cases} \frac{1}{9}\, t\, \mathbf{e}^{-\frac{t}{3}} & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

   (a) Verify that $f(t)$ is a probability density function.

   (b) What is the probability of a successful outcome happening in less than 12 minutes?

   (c) What is the probability of a successful outcome happening after 25 minutes?

   (d) What is the probability of a successful outcome happening between 10 and 75 minutes?

   (e) What is the mean time of a successful outcome from the experiment?

6. Determine the value of c for which the function below will be a probability density function.

$$f(x) = \begin{cases} c\left(12x^4 - x^5\right) & \text{if } 0 \leq x \leq 12 \\ 0 & \text{otherwise} \end{cases}$$

7. Use the function below for this problem and assume $a > 0$.

$$f(x) = \begin{cases} c\,\mathbf{e}^{-\frac{1}{a}x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

   (a) Determine the value of c for which this function will be a probability density function.

   (b) Using the value of c found in the first part determine the mean value of the probability density function.

9 Parametric Equations and Polar Coordinates

We are now going to take a look at a couple of topics that are completely different from anything we've seen to this point. That does not mean, however, that We can just forget everything that we've seen to this point. As we will see before too long we will still need to be able to do a large part of the material (both Calculus I and Calculus II material) that we've looked at to this point.

The first major topic that we'll look at in this chapter will be that of Parametric Equations. Parametric Equations will allow us to work with and perform Calculus operations on equations that cannot be solved into the form $y = f(x)$ or $x = h(y)$ (assuming we are using x and y as our variables). Also, as we'll see we can write some equations that can be solved for y or x as a set of easier to work with parametric equations.

Once we've got an idea of what parametric equations are and how to sketch graphs of them we will revisit some of the Calculus topics we've looked at to this point. We'll take a look at how to use only parametric equations to get the equation of tangent lines, where the graph is increasing/decreasing and the concavity of the graph. In addition, we'll revisit the idea of using a definite integral to find the area between the graph of a set of parametric equation and the x-axis. We will close out the Calculus topics by discussing arc length and surface area for a set of parametric equations.

We will then move into the other major topic of this chapter, namely Polar Coordinates. Once we've defined polar coordinates and gotten comfortable with them we will, again, revisit the same Calculus topics we looked at in terms of parametric equations.

On the surface it will appear that polar coordinates has nothing in common with parametric equations. We will see however that several topics in Polar Coordinates can be easily done, in some way, if we first set them up in terms of parametric equations.

In addition, we should point out that the purpose of the topics in this chapter is in preparation for multi-variable Calculus (*i.e.* the material that is usually taught in Calculus III). As we will see when we get to that point there are a lot of topics that involve and/or require parametric equations. In addition, polar coordinates will pop up every so often so keep that in mind as we go through this stuff. It is easy sometimes to get the idea that the topics in this chapter don't have a lot of use but once we hit multi-variable Calculus they will start to pop up with some regularity.

If you are looking for some practice problems (with solutions available) please check out the Prac-

tice Problems. There you will find a set of problems that should give you quite a bit practice.

9.1 Parametric Equations and Curves

For problems 1 - 9 eliminate the parameter for the given set of parametric equations, sketch the graph of the parametric curve and give any limits that might exist on x and y.

1. $x = 2 + t \qquad y = t^2 - 4t + 7$
2. $x = 2 + t \qquad y = t^2 - 4t + 7 \qquad -3 \leq t \leq 1$
3. $x = 1 - t^2 \qquad y = 3 + 2t$
4. $x = 1 - t^2 \qquad y = 3 + 2t \qquad -2 \leq t \leq 3$
5. $x = \frac{1}{2}\sqrt{t} \qquad y = t - \sqrt{t} - 6 \qquad t \geq 0$
6. $x = \frac{1}{2}\sqrt{t} \qquad y = t - \sqrt{t} - 6 \qquad 8 \leq t \leq 20$
7. $x = -6\cos(4t) \qquad y = 2\sin(4t) \qquad -\frac{\pi}{4} \leq t \leq \frac{\pi}{8}$
8. $x = 1 - 3\sin\left(\frac{1}{2}t\right) \qquad y = 4\cos\left(\frac{1}{2}t\right)$
9. $x = 6 - 7\mathbf{e}^{-2t} \qquad y = 4 + 3\mathbf{e}^{-2t}$
10. Answer each of the questions about the following set of parametric equations

$$x = 3\cos(at) \qquad y = 3\sin(at) \qquad 0 \leq t \leq 2\pi$$

(a) Sketch the graph of the parametric curve for $a = 1$.

(b) Sketch the graph of the parametric curve for $a = 6$.

(c) Sketch the graph of the parametric curve for $a = \frac{1}{5}$.

(d) In general, for $a > 0$, how does the value of a affect the graph of the parametric curve?

For problems 11 - 21 the path of a particle is given by the set of parametric equations. Completely describe the path of the particle. To completely describe the path of the particle you will need to provide the following information.

(i) A sketch of the parametric curve (including direction of motion) based on the equation you get by eliminating the parameter.

(ii) Limits on x and y.

(iii) A range of t's for a single trace of the parametric curve.

(iv) The number of traces of the curve the particle makes if an overall range of t's is provided in the problem.

11. $x = 6\cos\left(\frac{1}{3}t\right) \qquad y = 2 + \sin\left(\frac{1}{3}t\right) \qquad 0 \le t \le 75\pi$

12. $x = 7 - 3\sin(2t) \qquad y = 4 + 2\cos(2t)$

13. $x = 6\cos^2(3t) \qquad y = 2 - 3\sin^2(3t) \qquad -\frac{5}{6}\pi \le t \le 3\pi$

14. $x = \sqrt{2 + \cos^2\left(\frac{7}{4}t\right)} \qquad y = \frac{1}{3}\sin\left(\frac{7}{4}t\right)$

15. $x = 6 - \sin^3(4t) \qquad y = 2\sin(4t) \qquad -127\pi \le t \le 201\pi$

16. $x = 3 + \cos\left(\frac{1}{6}t\right) \qquad y = 4 + \cos^2\left(\frac{1}{6}t\right) \qquad -90\pi \le t \le 216\pi$

17. $x = \mathbf{e}^{-4t} \qquad y = 2\mathbf{e}^{12t}$

18. $x = 1 + \mathbf{e}^{3t} \qquad y = \mathbf{e}^{6t} \qquad -1 \le t \le 6$

19. $x = 1 - \ln(t) \qquad y = [\ln(t)]^2 \qquad t > 0$

20. $x = \cos\left(\frac{1}{2}t\right) \qquad y = \sec\left(\frac{1}{2}t\right) \qquad -\pi < t < \pi$

21. $x = \sin(2t) \qquad y = \sin^2(2t) - 4\sin(2t) \qquad -\frac{91}{4}\pi \le t \le \frac{17}{4}\pi$

For problems 22 - 27 write down a set of parametric equations for the given equation that meets the given extra conditions (if any).

22. $x = \sin\left(3 - y^2\right) + \cos^2(y)$

23. $y = \dfrac{6\cos(x) - 8}{x^2 + 9x}$

24. $x^2 + y^2 = 100$ and the parametric curve resulting from the parametric equations should be at $(0, 10)$ when $t = 0$ and the curve should have a clockwise rotation.

25. $x^2 + y^2 = 100$ and the parametric curve resulting from the parametric equations should be at $(0, 10)$ when $t = 0$ and the curve should have a counter clockwise rotation.

26. $\dfrac{x^2}{25} + y^2 = 1$ and the parametric curve resulting from the parametric equations should be at $(-5, 0)$ when $t = 0$ and the curve should have a counter clockwise rotation.

27. $\dfrac{x^2}{25} + y^2 = 1$ and the parametric curve resulting from the parametric equations should be at $(-5, 0)$ when $t = 0$ and the curve should have a clockwise rotation.

28. Eliminate the parameter for the following set of parametric equations and identify the resulting equation.

$$x = h + a\cos(\omega t) \qquad y = k + b\sin(\omega t)$$

9.2 Tangents with Parametric Equations

For problems 1 - 3 compute $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ for the given set of parametric equations.

1. $x = 7t^2 - 9t \qquad y = t^6 + 2t^2$
2. $x = \tan(2t) - 12 \qquad y = 3\sin(2t) + \sec(2t) + 4t$
3. $x = \ln\left(3t^2\right) + 8t \qquad y = \ln\left(t^4\right) - 6\ln\left(t^2\right)$

For problems 4 - 7 find the equation of the tangent line(s) to the given set of parametric equations at the given point.

4. $x = t^3 + \cos(\pi t) \qquad y = 4t + \sin(2t + 6)$ at $t = -3$
5. $x = t^2 + 2t - 1 \qquad y = t^3 + 7t^2 + 8t$ at $t = 1$
6. $x = 6 - \mathbf{e}^{t^3 - 9t} \qquad y = t^3 + 3t^2 - 18t + 2$ at $(5, 2)$
7. $x = t^2 + 5t - 6 \qquad y = t^2 + 2t - 8$ at $(-6, 7)$

For problems 8 and 9 find the values of t that will have horizontal or vertical tangent lines for the given set of parametric equations.

8. $x = t^3 - 5t^2 + t + 1 \qquad y = t^4 + 8t^3 + 3t^2$
9. $x = 7t^2 + \mathbf{e}^{2 - t^2} \qquad y = 10\sin\left(\dfrac{1}{2}t\right) - 1$

9.3 Area with Parametric Equations

For problems 1 - 3 determine the area of the region below the parametric curve given by the set of parametric equations. For each problem you may assume that each curve traces out exactly once from right to left for the given range of t. For these problems you should only use the given parametric equations to determine the answer.

1. $x = t^2 + 5t - 1 \qquad y = 40 - t^2 \qquad -2 \leq t \leq 5$
2. $x = 3\cos^2(t) - \sin^2(t) \qquad y = 6 + \cos(t) \qquad -\dfrac{\pi}{2} \leq t \leq 0$
3. $x = \mathbf{e}^{\frac{1}{4}t} - 2 \qquad y = 4 + \mathbf{e}^{\frac{1}{4}t} - \mathbf{e}^{\frac{1}{2}t} \qquad -6 \leq t \leq 1$

9.4 Arc Length with Parametric Equations

For all the problems in this section you should only use the given parametric equations to determine the answer.

For problems 1 - 5 determine the length of the parametric curve given by the set of parametric equations. For these problems you may assume that the curve traces out exactly once for the given range of t's.

1. $x = 3 + 9t \qquad y = 10 - 15t \qquad -5 \le t \le 8$
2. $x = 6(3+t)^{\frac{3}{2}} \qquad y = -3t^{\frac{3}{2}} \qquad -2 \le t \le 1$
3. $x = 4t^2 - 3 \qquad y = 3t \qquad 0 \le t \le 5$
4. $x = 3 + t \qquad y = 6 + (t-1)^2 \qquad 1 \le t \le 3$
5. $x = t^2 - 1 \qquad y = t^4 + 5 \qquad 0 \le t \le 1$

For problems 6 and 7 a particle travels along a path defined by the following set of parametric equations. Determine the total distance the particle travels and compare this to the length of the parametric curve itself.

6. $x = 6\cos^2(3t) \qquad y = 2 - 3\sin^2(3t) \qquad -\frac{5}{6}\pi \le t \le 3\pi$
7. $x = 3 + \cos\left(\frac{1}{6}t\right) \qquad y = 4 + \cos^2\left(\frac{1}{6}t\right) \qquad -90\pi \le t \le 216\pi$

For problems 8 - 10 set up, but do not evaluate, an integral that gives the length of the parametric curve given by the set of parametric equations. For these problems you may assume that the curve traces out exactly once for the given range of t's.

8. $x = t\cos(2t) \qquad y = \sin(3t) \qquad 2 \le t \le 3$
9. $x = 1 - \sin\left(1 + \sqrt{t}\right) \qquad y = \sin\left(\mathbf{e}^{-t}\right) \qquad 1 \le t \le 4$
10. $x = \ln(t+2) \qquad y = \frac{1}{t+7} \qquad -1 \le t \le 2$

9.5 Surface Area with Parametric Equations

For all the problems in this section you should only use the given parametric equations to determine the answer.

For problems1 - 4 determine the surface area of the object obtained by rotating the parametric curve about the given axis. For these problems you may assume that the curve traces out exactly once for the given range of t's.

1. Rotate $x = t^2 - 3 \qquad y = 2 + t^2 \qquad 0 \le t \le 5$ about the x-axis.
2. Rotate $x = -8t \qquad y = 6 + t^2 \qquad -3 \le t \le 0$ about the y-axis.
3. Rotate $x = t^2 \qquad y = t^4 - 2 \qquad 0 \le t \le 2$ about the y-axis.
4. Rotate $x = 2 + t \qquad y = 4\mathbf{e}^{-\frac{1}{2}t} \qquad -1 \le t \le 2$ about the x-axis.

For problems 5 - 7 set up, but do not evaluate, an integral that gives the surface area of the object obtained by rotating the parametric curve about the given axis. For these problems you may assume that the curve traces out exactly once for the given range of t's.

5. Rotate $x = 2 + \mathbf{e}^{\cos(t)} \qquad y = 1 + t^2 \qquad -2 \le t \le 0$ about the x-axis.
6. Rotate $x = \cos^2(t) \qquad y = 2\cos(2t) - \sin(t) \qquad 0 \le t \le 1$ about the y-axis.
7. Rotate $x = t^2 \qquad y = \ln\left(3 + \mathbf{e}^{-t}\right) \qquad 0 \le t \le 2$ about the x-axis.

9.6 Polar Coordinates

1. For the point with polar coordinates $\left(-9, \frac{3\pi}{7}\right)$ determine three different sets of coordinates for the same point all of which have angles different from $\frac{3\pi}{7}$ and are in the range $-2\pi \leq \theta \leq 2\pi$.

2. For the point with polar coordinates $\left(7, -\frac{2\pi}{3}\right)$ determine three different sets of coordinates for the same point all of which have angles different from $-\frac{2\pi}{3}$ and are in the range $-2\pi \leq \theta \leq 2\pi$.

3. The polar coordinates of a point are $(14,\ 2.48)$. Determine the Cartesian coordinates for the point.

4. The polar coordinates of a point are $\left(-\frac{3}{10},\ -5.29\right)$. Determine the Cartesian coordinates for the point.

5. The Cartesian coordinate of a point are $(-3, 5)$. Determine a set of polar coordinates for the point.

6. The Cartesian coordinate of a point are $(4, -7)$. Determine a set of polar coordinates for the point.

7. The Cartesian coordinate of a point are $(-3, -12)$. Determine a set of polar coordinates for the point.

For problems 8 and 9 convert the given equation into an equation in terms of polar coordinates.

8. $7x^2y + 8y = 3 - 6x^2 - 6y^2$

9. $\dfrac{7y}{x^2 + y^2 - 8x} = 9 + y^2$

For problems 10 - 13 convert the given equation into an equation in terms of Cartesian coordinates.

10. $r - \dfrac{8\sin(\theta)}{r} = 2\cos(\theta)$

11. $r^3 \csc(\theta) = 5\cos(\theta) - 6$

12. $8 - r = r^2 \sin(2\theta)$

13. $r = 2a\cos(\theta) + 2b\sin(\theta)$

For problems 14 - 27 sketch the graph of the given polar equation.

14. $-7 = r\sin(\theta)$
15. $\theta = \dfrac{5\pi}{7}$
16. $\theta = -\dfrac{9\pi}{5}$
17. $r\cos(\theta) = 4$
18. $r = 6\sin(\theta)$
19. $r = 100$
20. $r = 24\cos(\theta)$
21. $r = -15\sin(\theta)$
22. $r = 4 + 12\cos(\theta)$
23. $r = 7 - 7\sin(\theta)$
24. $r = 1 + 3\sin(\theta)$
25. $r = 5 - 4\cos(\theta)$
26. $r = 8 + 3\sin(\theta)$
27. $r = 1 - \cos(\theta)$

9.7 Tangents with Polar Coordinates

1. Find the tangent line to $r = \theta \sin(3\theta)$ at $\theta = \frac{\pi}{2}$.
2. Find the tangent line to $r = \cos(2\theta) - \sin(\theta)$ at $\theta = -\frac{\pi}{4}$.
3. Find the tangent line to $r = \cos\left(\theta^2 - \theta\right)$ at $\theta = \pi$.

9.8 Area with Polar Coordinates

1. Find the area inside the inner loop of $r = 3 + 10\sin(\theta)$.
2. Find the area inside the inner loop of $r = 5 + 12\cos(\theta)$.
3. Find the area inside the graph of $r = 8 + \cos(\theta)$ and to the right of the y-axis.
4. Find the area inside the graph of $r = 5 - 4\sin(\theta)$ and the below the x-axis.
5. Find the area that is inside $r = 4$ and outside $r = 4 - 2\sin(\theta)$.
6. Find the area that is inside $r = 7 - 3\cos(\theta)$ and outside $r = 4$.
7. Find the area that is inside $r = 6 + 6\cos(\theta)$ and outside $r = 4 - 3\cos(\theta)$.
8. Find the area that is inside $r = 4 + 2\sin(\theta)$ and outside $r = 5 - \sin(\theta)$.
9. Find the area that is inside $r = 5 - \sin(\theta)$ and outside $r = 4 + 2\sin(\theta)$.
10. Find the area that is inside both $r = 6 - 4\sin(\theta)$ and $r = 5$.
11. Find the area that is inside both $r = 3 + 2\cos(\theta)$ and $r = 3 - \cos(\theta)$.

9.9 Arc Length with Polar Coordinates

For problems 1 - 3 determine the length of the given polar curve. For these problems you may assume that the curve traces out exactly once for the given range of θ.

1. $r = \dfrac{1}{\cos(\theta)}$, $0 \leq \theta \leq \dfrac{\pi}{3}$
2. $r = \theta^2$, $0 \leq \theta \leq 3\pi$
3. $r = 6\cos(\theta) - 3\sin(\theta)$, $0 \leq \theta \leq \pi$

For problems 4 - 6 set up, but do not evaluate, an integral that gives the length of the given polar curve. For these problems you may assume that the curve traces out exactly once for the given range of θ.

4. $r = \sin\left(\theta^2\right)$, $0 \leq \theta \leq \pi$
5. $r = \cos\left(1 + \sin(\theta)\right)$, $0 \leq \theta \leq 2\pi$
6. $r = \mathbf{e}^{-\frac{1}{4}\theta}\cos(\theta)$, $0 \leq \theta \leq 3\pi$

9.10 Surface Area with Polar Coordinates

For problems 1 - 4 set up, but do not evaluate, an integral that gives the surface area of the curve rotated about the given axis. For these problems you may assume that the curve traces out exactly once for the given range of θ.

1. $r = \cos\left(\mathbf{e}^{-\frac{1}{4}\theta}\right)$, $0 \le \theta \le \dfrac{\pi}{2}$ rotated about the x-axis.

2. $r = \theta \sin(\theta)$, $0 \le \theta \le \dfrac{\pi}{2}$ rotated about the y-axis.

3. $r = \cos(\theta)\sin(2\theta)$, $0 \le \theta \le \dfrac{\pi}{6}$ rotated about the x-axis.

4. $r = \theta + \sin(\theta)$, $\dfrac{\pi}{2} \le \theta \le \pi$ rotated about the y-axis.

9.11 Arc Length and Surface Area Revisited

Problems have not yet been written for this section and probably won't be to be honest since this is just a summary section.

10 Series and Sequences

Once again, as with the last chapter, we are going to be looking at a completely different topic in this chapter. The only material from previous chapters that will be needed here will be the ability to compute limits at infinity (we'll do a fair amount of these), compute the rare derivatives and compute the occasional integral. The integrals will, generally, be fairly simple and needing u substitutions every once in a while although we will see the occasional integral requiring integration by parts or partial fractions. So, basically, the material in this chapter doesn't really rely all that much on previous material.

Series is one of those topics that many students don't find all that useful. To be honest, many students will never see series outside of their calculus class. However, series do play an important role in the field of ordinary differential equations and without series large portions of the field of partial differential equations would not be possible.

In other words, series is an important topic even if you won't ever see any of the applications. Most of the applications are beyond the scope of most Calculus courses and tend to occur in classes that many students don't take. So, as you go through this material keep in mind that these do have applications even if we won't really be covering many of them in this class.

The first topic we'll be looking at in this chapter is that of a sequence. We'll define just what we mean by a sequence and look at some basic topics and concepts that we'll need to work with them.

The other topic will be that of (infinite) series. In fact, we will spend the vast majority of this chapter deal with series. We can't, however, fully discuss series without understanding sequences and hence the reason for discussing sequences first. We will define just what an infinite series is and what it means for a series to converge or diverge. The majority of this chapter will then be spent discussing a variety of methods for testing whether or not a series will converge or diverge.

We'll close out the chapter with a discussion of power series and Taylor series as well as a couple of quick applications of series that we can easily discuss without needing any extra knowledge (as is needed for most applications of series).

If you are looking for some practice problems (with solutions available) please check out the Practice Problems. There you will find a set of problems that should give you quite a bit practice.

10.1 Sequences

For problems 1 - 3 list the first 5 terms of the sequence.

1. $\left\{(-1)^n \mathbf{e}^{2n}\right\}_{n=1}^{\infty}$
2. $\left\{\dfrac{6-8n}{n^2+9n}\right\}_{n=2}^{\infty}$
3. $\left\{\dfrac{4}{(-2)^{n+1}+3^n}\right\}_{n=1}^{\infty}$

For problems 4 - 10 determine if the given sequence converges or diverges. If it converges what is its limit?

4. $\left\{\dfrac{5+n^3}{2n^2-8n+1}\right\}_{n=0}^{\infty}$
5. $\left\{\dfrac{6n^4+9n^2}{9n^4-8n^2+7}\right\}_{n=11}^{\infty}$
6. $\left\{\dfrac{(-1)^{n+7}(2-8n)}{n^2+9}\right\}_{n=2}^{\infty}$
7. $\left\{\cos(n\pi)\right\}_{n=0}^{\infty}$
8. $\left\{\dfrac{n+1}{\ln(6n)}\right\}_{n=2}^{\infty}$
9. $\left\{\cos\left(\dfrac{3}{n+1}\right)\right\}_{n=1}^{\infty}$
10. $\left\{\ln(4n+1)-\ln(2+7n)\right\}_{n=0}^{\infty}$

10.2 More on Sequences

For each of the following problems determine if the sequence is increasing, decreasing, not monotonic, bounded below, bounded above and/or bounded.

1. $\left\{\dfrac{1}{n^3+1}\right\}_{n=1}^{\infty}$
2. $\left\{\mathbf{e}^{3n}\right\}_{n=0}^{\infty}$
3. $\left\{(-3)^n\right\}_{n=0}^{\infty}$
4. $\{\sin(n)\}_{n=4}^{\infty}$
5. $\left\{\ln\left(\dfrac{1}{n}\right)\right\}_{n=2}^{\infty}$
6. $\left\{\dfrac{3-n}{1-3n}\right\}_{n=1}^{\infty}$
7. $\left\{\dfrac{2n+1}{4n+3}\right\}_{n=0}^{\infty}$
8. $\left\{(1-n)\,\mathbf{e}^n\right\}_{n=3}^{\infty}$
9. $\left\{\dfrac{n^2+40}{n^2+3n+1}\right\}_{n=1}^{\infty}$
10. $\left\{\dfrac{5+n}{100,000+n^2}\right\}_{n=0}^{\infty}$

10.3 Series - Basics

For problems 1 - 4 perform an index shift so that the series starts at $n = 4$.

1. $\displaystyle\sum_{n=8}^{\infty} \frac{2+n}{5-n}$

2. $\displaystyle\sum_{n=2}^{\infty} \frac{3^{n+2}}{1-4^{3n+1}}$

3. $\displaystyle\sum_{n=0}^{\infty} (-2)^{2-n}\,\mathbf{e}^{3n}$

4. $\displaystyle\sum_{n=5}^{\infty} \frac{(-1)^{3+n}n^2}{n^2-2n+1}$

5. Strip out the first 4 terms from the series $\displaystyle\sum_{n=0}^{\infty} 3^n\,6^{2-n}$.

6. Strip out the first 2 terms from the series $\displaystyle\sum_{n=3}^{\infty} \frac{4}{n^2+n+1}$.

7. Given that $\displaystyle\sum_{n=4}^{\infty} n\,4^{-n} = 0.02257$ determine the value of $\displaystyle\sum_{n=1}^{\infty} n\,4^{-n}$.

8. Given that $\displaystyle\sum_{n=3}^{\infty} \frac{n+1}{n^3} = 0.47199$ determine the value of $\displaystyle\sum_{n=5}^{\infty} \frac{n+1}{n^3}$.

10.4 Convergence & Divergence of Series

For problems 1 - 4 compute the first 3 terms in the sequence of partial sums for the given series.

1. $\displaystyle\sum_{n=0}^{\infty} \frac{1}{1+3^n}$

2. $\displaystyle\sum_{n=1}^{\infty} (2^n - 3^n)$

3. $\displaystyle\sum_{n=1}^{\infty} \frac{1+n}{2n}$

4. $\displaystyle\sum_{n=0}^{\infty} 10$

For problems 5 - 7 assume that the nth term in the sequence of partial sums for the series $\displaystyle\sum_{n=0}^{\infty} a_n$ is given below. Determine if the series $\displaystyle\sum_{n=0}^{\infty} a_n$ is convergent or divergent. If the series is convergent determine the value of the series.

5. $s_n = \left(n^2 + 4n\right) \mathbf{e}^{-2n}$

6. $s_n = \dfrac{1 + 2n + 3n^2}{4n^2 + 5n + 6}$

7. $s_n = \dfrac{n}{\ln(n+2)}$

8. Let $d_n = \dfrac{7 - 8n}{4 + 3n}$

   (a) Does the sequence $\{d_n\}_{n=0}^{\infty}$ converge or diverge?

   (b) Does the series $\displaystyle\sum_{n=0}^{\infty} d_n$ converge or diverge?

9. Let $d_n = 1 + n\mathbf{e}^{-n}$

   (a) Does the sequence $\{d_n\}_{n=0}^{\infty}$ converge or diverge?

   (b) Does the series $\displaystyle\sum_{n=0}^{\infty} d_n$ converge or diverge?

For problems 10 - 12 show that the series is divergent.

10. $\displaystyle\sum_{n=1}^{\infty} \frac{9-2n^2}{1+4n+n^2}$

11. $\displaystyle\sum_{n=0}^{\infty} \frac{5^n+1}{3^n}$

12. $\displaystyle\sum_{n=1}^{\infty} \cos(n)$

10.5 Special Series

For each of the following series determine if the series converges or diverges. If the series converges give its value.

1. $\displaystyle\sum_{n=2}^{\infty} \frac{-2}{n^2+n}$

2. $\displaystyle\sum_{n=1}^{\infty} \frac{12}{n}$

3. $\displaystyle\sum_{n=1}^{\infty} 5^{n+3}\, 4^n$

4. $\displaystyle\sum_{n=0}^{\infty} \frac{3}{4^{n+1}\, 5^{1-n}}$

5. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{14\, n}$

6. $\displaystyle\sum_{n=0}^{\infty} \frac{7}{n^2+5n+6}$

7. $\displaystyle\sum_{n=1}^{\infty} 4^{1+2n}\, 3^{2-3n}$

8. $\displaystyle\sum_{n=4}^{\infty} 4^{1+2n}\, 3^{2-3n}$

9. $\displaystyle\sum_{n=3}^{\infty} \frac{5}{n^2-1}$

10. $\displaystyle\sum_{n=4}^{\infty} \frac{1}{n^2-4n+3}$

11. $\displaystyle\sum_{n=0}^{\infty} \frac{5^{3+n}}{2^{2+3n}}$

10.6 Integral Test

For each of the following series determine if the series converges or diverges.

1. $\displaystyle\sum_{n=2}^{\infty} \frac{4}{(\sqrt{n})^3}$

2. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt[7]{n^2}\ \sqrt[6]{n}}$

3. $\displaystyle\sum_{n=2}^{\infty} \frac{1}{2n+1}$

4. $\displaystyle\sum_{n=0}^{\infty} \frac{8}{(n+10)^2}$

5. $\displaystyle\sum_{n=0}^{\infty} \frac{1}{n^2+1}$

6. $\displaystyle\sum_{n=2}^{\infty} \frac{\ln(n)}{n}$

7. $\displaystyle\sum_{n=0}^{\infty} \frac{n^3}{n^4+1}$

8. $\displaystyle\sum_{n=0}^{\infty} \frac{n^3}{(n^4+1)^2}$

9. $\displaystyle\sum_{n=4}^{\infty} \frac{4}{n^2-n-6}$

10. $\displaystyle\sum_{n=1}^{\infty} \frac{9}{n^2+5n+4}$

11. $\displaystyle\sum_{n=0}^{\infty} n\,\mathbf{e}^{-n}$

10.7 Comparison & Limit Comparison Test

For each of the following series determine if the series converges or diverges.

1. $\displaystyle\sum_{n=0}^{\infty} \frac{3^n + n}{2^{n+1}}$

2. $\displaystyle\sum_{n=1}^{\infty} \frac{4n - 3}{2n^5}$

3. $\displaystyle\sum_{n=4}^{\infty} \frac{1}{(2n-1)(n-3)}$

4. $\displaystyle\sum_{n=8}^{\infty} \frac{\ln\left(n^2\right)}{n}$

5. $\displaystyle\sum_{n=1}^{\infty} \frac{4n}{(n+1)^3}$

6. $\displaystyle\sum_{n=0}^{\infty} \frac{n-4}{(n^2+1)\,\mathbf{e}^n}$

7. $\displaystyle\sum_{n=2}^{\infty} \frac{\sqrt{2 + \cos^2(5n)}}{\sqrt{n^2 - n - 1}}$

8. $\displaystyle\sum_{n=3}^{\infty} \frac{n-1}{\sqrt{n^3 + n + 3}}$

9. $\displaystyle\sum_{n=0}^{\infty} \frac{3n^2 + 7n - 1}{n^4 - n + 3}$

10. $\displaystyle\sum_{n=1}^{\infty} \frac{(1 - \sin(n))(1 + \sin(n))}{n^2 + 8n + 1}$

10.8 Alternating Series Test

For each of the following series determine if the series converges or diverges.

1. $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^{n+7}}{n^2+3}$

2. $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^{n-2}}{3^n+3n}$

3. $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^{n-1}}{n^3+4n^2+8}$

4. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(-2)^n(6n+1)}$

5. $\displaystyle\sum_{n=3}^{\infty} \frac{4n\cos(n\pi)}{2n^2+1}$

6. $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^{n-10}\,n^2}{n^3+n^2+4}$

7. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+5}(2n+1)}{n^2+8}$

10.9 Absolute Convergence

For each of the following series determine if they are absolutely convergent or conditionally convergent.

1. $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^{n-2}}{\sqrt[3]{n-1}}$

2. $\displaystyle\sum_{n=3}^{\infty} \frac{\cos(n\pi)}{n^4}$

3. $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^{n-3} n}{4n^2+3}$

4. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+6}\left(1+n^2\right)}{n^4}$

5. $\displaystyle\sum_{n=2}^{\infty} \frac{\cos^3(n)}{n^3-n}$

10.10 Ratio Test

For each of the following series determine if the series converges or diverges.

1. $\displaystyle\sum_{n=0}^{\infty} \frac{n^3+n^2}{(n+1)!}$

2. $\displaystyle\sum_{n=1}^{\infty} \frac{n+2}{5^{1-n}\,(n+1)}$

3. $\displaystyle\sum_{n=0}^{\infty} \frac{(2n-1)!}{(3n)!}$

4. $\displaystyle\sum_{n=0}^{\infty} \frac{(-2)^{4+n}}{3n^2+1}$

5. $\displaystyle\sum_{n=2}^{\infty} \frac{4^{1+\frac{1}{2}n}n^2}{3^{2+n}\,(n+3)}$

6. $\displaystyle\sum_{n=1}^{\infty} \frac{4}{(-1)^{n+2}\,(n^2+n+1)}$

7. $\displaystyle\sum_{n=3}^{\infty} \frac{6^{-2n}\,(n-4)}{4^{3-2n}\,(2-n^2)}$

8. $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^n\,(n+1)}{n^2+1}$

10.11 Root Test

For each of the following series determine if the series converges or diverges.

1. $\displaystyle\sum_{n=0}^{\infty} \frac{2^{1-3n}}{(-3)^{1-2n}}$

2. $\displaystyle\sum_{n=2}^{\infty} \left(\frac{5n^2-2n+1}{3n^2+n-3}\right)^{-n}$

3. $\displaystyle\sum_{n=1}^{\infty} \frac{10^{3+5n}}{n^{1+n^2}}$

4. $\displaystyle\sum_{n=1}^{\infty} \left(\frac{6-9n}{2+4n}\right)^{\frac{1}{3}n}$

10.12 Strategy for Series

Problems have not yet been written for this section.

I was finding it very difficult to come up with a good mix of new problems and decided my time was better spent writing problems for later sections rather than trying to come up with a sufficient number of problems for what is essentially a review section. I intend to come back at a later date when I have more time to devote to this section and add problems then.

10.13 Estimating the Value of a Series

1. Use the Integral Test and $n = 8$ to estimate the value of $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n\,(\ln(n))^2}$.

2. Use the Integral Test and $n = 14$ to estimate the value of $\displaystyle\sum_{n=0}^{\infty} \frac{1}{(2n+1)^{\frac{5}{2}}}$.

3. Use the Comparison Test and $n = 10$ to estimate the value of $\displaystyle\sum_{n=0}^{\infty} \frac{3^n - 2}{2^{2n} + 1}$.

4. Use the Comparison Test and $n = 8$ to estimate the value of $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{n^4 + 1}$.

5. Use the Alternating Series Test and $n = 12$ to estimate the value of $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{n+1}$.

6. Use the Alternating Series Test and $n = 18$ to estimate the value of $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n}}{3n+4}$.

7. Use the Ratio Test and $n = 10$ to estimate the value of $\displaystyle\sum_{n=1}^{\infty} \frac{(-2)^{1+2n}}{n^2\, 7^n}$.

8. Use the Ratio Test and $n = 5$ to estimate the value of $\displaystyle\sum_{n=1}^{\infty} \frac{n}{(n-1)!}$.

10.14 Power Series

For each of the following power series determine the interval and radius of convergence.

1. $\displaystyle\sum_{n=0}^{\infty} \frac{6^{1-n}}{(-2)^{3-2n}}(x+4)^n$

2. $\displaystyle\sum_{n=1}^{\infty} \frac{(10x-1)^n}{n^{3+n}}$

3. $\displaystyle\sum_{n=0}^{\infty} \frac{(3n)!}{(2n-2)!}(6x-9)^n$

4. $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n n^2}{4n+1}(5x+20)^n$

5. $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n\, 8^{1+n}}{n+4}(x-7)^n$

6. $\displaystyle\sum_{n=1}^{\infty} \frac{2^{1+2n}}{(-3)^{1+2n}\, n^2}(4x+2)^n$

7. $\displaystyle\sum_{n=0}^{\infty} \frac{(x+12)^{2+n}}{(-16)^{5+n}}$

10.15 Power Series and Functions

For problems 1 - 4 write the given function as a power series and give the interval of convergence.

1. $f(x) = \dfrac{x}{1-8x}$

2. $f(x) = \dfrac{-12x^2}{1+6x^7}$

3. $f(x) = \dfrac{x^7}{8+x^3}$

4. $f(x) = \dfrac{\sqrt[5]{x^2}}{4-3x^2}$

For problems 5 & 6 give a power series representation for the derivative of the following function.

5. $g(x) = \dfrac{x^{10}}{2-x^2}$

6. $g(x) = \dfrac{9x^5}{1+3x^6}$

For problems 7 & 8 give a power series representation for the integral of the following function.

7. $h(x) = \dfrac{7x}{3-6x}$

8. $h(x) = \dfrac{x^4}{2+x^9}$

10.16 Taylor Series

For problems 1 - 3 use one of the Taylor Series derived in the notes to determine the Taylor Series for the given function.

1. $f(x) = \sin\left(x^4\right)$ about $x = 0$
2. $f(x) = 9x^4\mathbf{e}^{-12x}$ about $x = 0$
3. $f(x) = 6x^2\cos\left(7x^5\right)$ about $x = 0$

For problem 4 - 13 find the Taylor Series for each of the following functions.

4. $f(x) = \sin(x)$ about $x = \dfrac{3\pi}{2}$
5. $f(x) = \mathbf{e}^{1-8x}$ about $x = 3$
6. $f(x) = \ln(1-x)$ about $x = -2$
7. $f(x) = \ln(2+9x)$ about $x = 1$
8. $f(x) = \dfrac{1}{(6-x)^7}$ about $x = 4$
9. $f(x) = \dfrac{1}{(4+9x)^2}$ about $x = -2$
10. $f(x) = \sqrt{2+x}$ about $x = 1$
11. $f(x) = \sqrt{1-4x}$ about $x = -3$
12. $f(x) = -3x^2 - x + 10$ about $x = -8$
13. $f(x) = x^3 + 9x^2 - 10x + 2$ about $x = 3$

10.17 Applications of Series

1. Determine a Taylor Series about $x = 0$ for the following integral.

$$\int \frac{\cos(x) - 1}{x}\, dx$$

2. Write down $T_2(x)$, $T_4(x)$ and $T_6(x)$ for the Taylor Series of $f(x) = \sin(x)$ about $x = \frac{3\pi}{2}$. Graph all three of the Taylor polynomials and $f(x)$ on the same graph for the interval $[-\pi, 2\pi]$.

3. Write down $T_2(x)$, $T_3(x)$ and $T_4(x)$ for the Taylor Series of $f(x) = \ln(1 - x)$ about $x = -2$. Graph all three of the Taylor polynomials and $f(x)$ on the same graph for the interval $[-4, 0]$.

4. Write down $T_1(x)$, $T_3(x)$ and $T_5(x)$ for the Taylor Series of $f(x) = \frac{1}{(6-x)^7}$ about $x = 4$. Graph all three of the Taylor polynomials and $f(x)$ on the same graph for the interval $[1, 5]$.

5. Write down $T_2(x)$, $T_4(x)$ and $T_6(x)$ for the Taylor Series of $f(x) = \sqrt{2 + x}$ about $x = 1$. Graph all three of the Taylor polynomials and $f(x)$ on the same graph for the interval $[-2, 4]$.

10.18 Binomial Series

For problems 1 - 3 use the Binomial Theorem to expand the given function.

1. $\left(\frac{1}{2}+\frac{2}{3}x\right)^4$
2. $(1-8x)^5$
3. $(3+7x)^6$

For problems 4 - 6 write down the first four terms in the binomial series for the given function.

4. $(8+4x)^{-4}$
5. $\sqrt{16+25x}$
6. $\sqrt[4]{16-4x}$

11 Vectors

Once again we are completely changing topics from the last chapter. We are going to do a (very) brief introduction to vectors. We'll look at basic notation and concepts involving vectors as well as arithmetic involving vectors. We'll also look at the dot product and cross product of vectors as well as a couple of quick applications of the dot and cross product.

Once we get into the multi-variable Calculus (*i.e.* the topics usually taught in Calculus III) we'll run into vectors on a semi regular basis and so we'll need to be familiar with them and the common notation, concepts and arithmetic involving vectors.

If you are looking for some practice problems (with solutions available) please check out the Practice Problems. There you will find a set of problems that should give you quite a bit practice.

11.1 Basic Concepts

1. Describe the difference between $(4, -1)$ and $\langle 4, -1 \rangle$. Illustrate the difference with a sketch.

For problems 2 - 6 give the vector for the set of points. Find its magnitude and determine if the vector is a unit vector.

2. The line segment from $(6, -2, 3)$ to $(-3, -2, 1)$.
3. The line segment from $\left(1, -\frac{1}{5}\right)$ to $\left(\frac{8}{5}, \frac{3}{5}\right)$.
4. The line segment from $(2, -1, 5)$ to $(8, -3, -6)$.
5. The position vector for $(-12, 4, 8)$.
6. The position vector for $(\cos(\theta), \sin(\theta))$ for any angle θ.
7. The vector $\vec{v} = \langle -8, -3 \rangle$ starts at the point $P = (8, 2)$. At what point does the vector end?
8. The vector $\vec{v} = \langle 0, 5, -3 \rangle$ starts at the point $P = (-1, 0, 5)$. At what point does the vector end?
9. The vector $\vec{v} = \langle -8, -3 \rangle$ ends at the point $P = (8, 2)$. At what point does the vector start?
10. The vector $\vec{v} = \langle 4, -2, 1 \rangle$ ends at the point $P = (7, -7, 2)$. At what point does the vector start?

11.2 Vector Arithmetic

1. Given $\vec{a} = 3\vec{i} - 9\vec{j}$ and $\vec{b} = -6\vec{i} + \vec{j}$ compute each of the following.
   (a) $10\vec{b}$
   (b) $14\vec{a} + 20\vec{b}$
   (c) $\left\| 8\vec{b} - \frac{1}{3}\vec{a} \right\|$
2. Given $\vec{u} = \langle 0, 4, -1 \rangle$ and $\vec{v} = \langle 6, -2, -7 \rangle$ compute each of the following.
   (a) $\frac{3}{4}\vec{u}$
   (b) $-3\vec{u} - 7\vec{v}$
   (c) $\|\vec{v} + 10\vec{u}\|$
3. Given $\vec{p} = \langle 3, -1, -2 \rangle$ and $\vec{q} = -\frac{1}{3}\vec{i} - \frac{1}{2}\vec{k}$ compute each of the following.
   (a) $2\vec{p}$
   (b) $9\vec{q} - 2\vec{p}$
   (c) $\|8\vec{p} - 12\vec{q}\|$
4. Find a unit vector that points in the same direction as $\vec{a} = \langle 10, -3, 8, -2 \rangle$.
5. Find a unit vector that points in the same direction as $\vec{w} = -\vec{i} - 6\vec{j}$.
6. Find a unit vector that points in the opposite direction as $\vec{c} = 2\vec{i} + 7\vec{j} - 5\vec{k}$.
7. Find a unit vector that points in the opposite direction as $\vec{b} = \langle 0, -3, -11 \rangle$.
8. Find a vector that points in the same direction as $\vec{p} = 2\vec{i} - 3\vec{j} + \vec{k}$ with a magnitude of $\frac{1}{2}$.
9. Find a vector that points in the opposite direction as $\vec{a} = \langle -3, -14, 2 \rangle$ with a magnitude of 32.
10. Find a vector that points in the same direction as $\vec{b} = -3\vec{i} + 2\vec{k}$ with a magnitude that is $\frac{1}{10}$ the magnitude of $\vec{b}$.
11. Determine if $\vec{p} = 8\vec{i} - 3\vec{j}$ and $\vec{q} = 16\vec{i} - 6\vec{j}$ are parallel vectors.
12. Determine if $\vec{v} = \langle 1, 0, -4 \rangle$ and $\vec{w} = \langle 9, 3, 1 \rangle$ are parallel vectors.
13. Determine if $\vec{a} = 10\vec{i} + 8\vec{j} + 20\vec{k}$ and $b = \langle -35, -28, 70 \rangle$ are parallel vectors.

14. Prove the property : $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$.

15. Prove the property : $\vec{v} + \vec{0} = \vec{v}$.

16. Prove the property : $1\vec{v} = \vec{v}$.

17. Prove the property : $(a + b)\,\vec{v} = a\vec{v} + b\vec{v}$.

11.3 Dot Product

For problems 1 - 5 determine the dot product, $\vec{a} \cdot \vec{b}$.

1. $\vec{a} = 9\vec{i} - 8\vec{k}$, $\vec{b} = \langle 3, -2, 1 \rangle$
2. $\vec{a} = \langle 4, -1, 0, 5 \rangle$, $\vec{b} = \langle 3, 0, -10, 6 \rangle$
3. $\vec{a} = \vec{i} - 5\vec{j} - 2\vec{k}$, $\vec{b} = -4\vec{i} + 2\vec{j} + 8\vec{k}$
4. $\|\vec{a}\| = \dfrac{1}{4}$, $\left\|\vec{b}\right\| = \dfrac{9}{4}$ and the angle between the two vectors is $\theta = \pi$.
5. $\|\vec{a}\| = 24$, $\left\|\vec{b}\right\| = 9$ and the angle between the two vectors is $\theta = \dfrac{2\pi}{7}$.

For problems 6 - 8 determine the angle between the two vectors.

6. $\vec{p} = 9\vec{i} - \vec{j}$, $\vec{q} = -3\vec{i} - 6\vec{j}$
7. $\vec{a} = \langle 4, 0, -3 \rangle$, $\vec{b} = 2\vec{i} + 10\vec{j} - 11\vec{k}$
8. $\vec{w} = \langle 8, 3, -1, -4 \rangle$, $\vec{v} = \langle -1, 9, 4, -8 \rangle$

For problems 9 - 12 determine if the two vectors are parallel, orthogonal or neither.

9. $\vec{q} = 7\vec{i} - 14\vec{j} - 21\vec{k}$, $\vec{p} = \langle -4, 8, 12 \rangle$
10. $\vec{u} = \langle 5, 0, -2 \rangle$, $\vec{q} = \langle 4, -7, 10 \rangle$
11. $\vec{a} = 9\vec{i} - \vec{j} + 5\vec{k}$, $\vec{b} = -2\vec{i} + 7\vec{j} + \vec{k}$
12. $\vec{v} = \langle -1, 3, 1, 5 \rangle$, $\vec{w} = \langle -8, 3, -7, -2 \rangle$
13. Given that $\vec{a} \cdot \vec{b} = -6$, $\|\vec{a}\| = 4.3$ and the angle between $\vec{a}$ and $\vec{b}$ is $\theta = \dfrac{\pi}{6}$ determine if $\vec{b}$ is a unit vector or not.

For problems 14 & 15 determine the value of b for which the two vectors will be orthogonal.

14. $\vec{u} = \langle 3, -1, 6 \rangle$, $\vec{v} = \langle 3, -2b, 1 \rangle$
15. $\vec{u} = \langle 1 - b, 4, -2 \rangle$, $\vec{v} = \langle b, 6, 3b \rangle$
16. Given $\vec{a} = \vec{i} + 3\vec{j} - 2\vec{k}$ and $\vec{b} = -3\vec{i} - 4\vec{j} + 7\vec{k}$ compute $\text{proj}_{\vec{a}}\,\vec{b}$.
17. Given $\vec{a} = \vec{i} + 3\vec{j} - 2\vec{k}$ and $\vec{b} = -3\vec{i} - 4\vec{j} + 7\vec{k}$ compute $\text{proj}_{\vec{b}}\,\vec{a}$.

18. Given $\vec{p} = \langle 5, -2, 1 \rangle$ and $\vec{q} = \langle 0, 4, 8 \rangle$ compute $\text{proj}_{\vec{p}}\,\vec{q}$.

19. Given $\vec{u} = \langle 1, 3, 0, -2 \rangle$ and $\vec{w} = \langle -2, 2, 4, 1 \rangle$ compute $\text{proj}_{\vec{w}}\,\vec{u}$.

20. Determine the direction cosines and direction angles for $\vec{r} = \langle 5, 2, -7 \rangle$.

21. Determine the direction cosines and direction angles for $\vec{r} = \left\langle \frac{1}{2}, -\frac{3}{4}, \frac{5}{2} \right\rangle$.

22. Prove the property $(c\vec{v}) \cdot \vec{w} = \vec{v} \cdot (c\vec{w})$.

23. Prove the property $\vec{v} \cdot \vec{w} = \vec{w} \cdot \vec{v}$.

24. Prove the property $\vec{v} \cdot \vec{0} = 0$.

25. Prove the property $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2$.

11.4 Cross Product

1. If $\vec{w} = \langle 1, 0, -3 \rangle$ and $\vec{v} = \langle 6, -3, -4 \rangle$ compute $\vec{v} \times \vec{w}$.

2. If $\vec{w} = \langle 1, 0, -3 \rangle$ and $\vec{v} = \langle 6, -3, -4 \rangle$ compute $\vec{w} \times \vec{v}$.

3. If $\vec{a} = 3\vec{i} - 2\vec{j} + 6\vec{k}$ and $\vec{b} = \langle 4, -1, -6 \rangle$ compute $\vec{a} \times \vec{b}$.

4. Find a vector that is orthogonal to the plane containing the points $P = (-4, 2, 6)$, $Q = (-3, 2, 1)$ and $R = (2, -1, 1)$.

5. Find a vector that is orthogonal to the plane containing the points $P = (-1, 1, 6)$, $Q = (-2, 3, 2)$ and $R = (-2, 4, 5)$.

6. Are the vectors $\vec{u} = \langle -2, 4, -1 \rangle$, $\vec{v} = \langle 5, -2, -1 \rangle$ and $\vec{w} = \langle 3, 4, -3 \rangle$ are in the same plane?

7. Are the vectors $\vec{u} = \langle 1, -1, 4 \rangle$, $\vec{v} = \langle 4, 2, -2 \rangle$ and $\vec{w} = \langle -5, 4, -17 \rangle$ are in the same plane?

8. Determine the value of b so that the vectors $\vec{u} = \langle 4, -5, 3 \rangle$, $\vec{v} = \langle -2, 0, -5 \rangle$ and $\vec{w} = \langle b, -1, 6 \rangle$ are in the same plane.

12 Three Dimensional Space

In this chapter we will start looking at three dimensional space (3-D space or $\mathbb{R}^3$).As with the last chapter this is preparation for multi-variable Calculus (which we'll be starting in the next chapter) as the vast majority of the multi-variable Calculus material assumes we are in three dimensional (or higher dimensional) space.

In this chapter we will discuss the equations of lines and planes in three dimensional space as well as the equations of many of the standard quadric surfaces (*i.e* equations with at least one quadratic term in it).

We will define a vector function and discuss how to perform basic Calculus operations on vector functions. We will also discuss how to get tangent vectors (a vector tangent to a curve), normal vectors (a vector orthogonal/perpendicular) and the curvature of a curve from the vector function that defines the curve. We'll also have a quick discussion of how to get the velocity and acceleration of an object as it travels along a curve defined by a vector function.

We will close out the chapter with a discussion a couple of alternative coordinates systems for three dimensional space, namely, cylindrical coordinates (a 3D extension of polar coordinates) and spherical coordinates.

If you are looking for some practice problems (with solutions available) please check out the Practice Problems. There you will find a set of problems that should give you quite a bit practice.

12.1 The 3-D Coordinate System

1. Give the projection of $P = (-9, 1, 5)$ onto the three coordinate planes.
2. Give the projection of $P = (3, -2, -5)$ onto the three coordinate planes.
3. Which of the points $P = (8, -9, 3)$ and $Q = (-6, 4, -5)$ is closest to the xz-plane?
4. Which of the points $P = (8, -9, 3)$ and $Q = (-6, 4, -5)$ is closest to the xy-plane?
5. Which of the points $P = (5, -4, 3)$ and $Q = (-6, 3, 9)$ is closest to the x-axis?
6. Which of the points $P = (5, -4, 3)$ and $Q = (-6, 3, 9)$ is closest to the y-axis?

For problems 7 - 9 list all of the coordinates systems ($\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^3$) that the given equation will have a graph in. Do not sketch the graph.

7. $8z + \dfrac{x+1}{y^2+2} = 4x$
8. $\sqrt{y+2} = 6$
9. $7y^3 - \dfrac{2}{x+1} = xy$

12.2 Equations of Lines

For problems 1 - 4 give the equation of the line in vector form, parametric form and symmetric form.

1. The line through the points $(7, -3, 1)$ and $(-2, 1, 4)$.
2. The line through the point $(1, -5, 0)$ and parallel to the line given by $\vec{r}(t) = \langle 8 - 3t, -10 + 9t, -1 - t\rangle$.
3. The line through the point $(1, -7, 14)$ and parallel to the line given by $x = 6t$, $y = 9$, $z = 8-16t$.
4. The line through the point $(-7, 2, 4)$ and orthogonal to both $\vec{v} = \langle 0, -9, 1\rangle$ and $\vec{w} = 3\vec{i}+\vec{j}-4\vec{k}$.

For problems 5 - 7 determine if the two lines are parallel, orthogonal or neither.

5. The line given by $\vec{r}(t) = \langle 4 - 7t, -10 + 5t, 21 - 4t\rangle$ and the line given by $\vec{r}(t) = \langle -2 + 3t, 7 + 5t, 5 + t\rangle$.
6. The line through the points $(10, -4, 18)$ and $(5, 6, -7)$ and the line given by $x = 5+3t$, $y = -6t$, $z = 1 + 15t$.
7. The line given by $x = 29$, $y = -3 - 6t$, $z = 12 - t$ and the line given by $\vec{r}(t) = \langle 12 - 14t, 2 + 7t, -10 + 3t\rangle$.

For problems 8 - 10 determine the intersection point of the two lines or show that they do not intersect.

8. The line passing through the points $(0, -9, -1)$ and $(1, 6, -3)$ and the line given by $\vec{r}(t) = \langle -9 - 4t, 10 + 6t, 1 - 2t\rangle$.
9. The line given by $x = 1 + 6t$, $y = -1 - 3t$, $z = 4 + 12t$ and the line given by $x = 4 + t$, $y = -10 - 8t$, $z = 3 - 5t$.
10. The line given by $\vec{r}(t) = \langle 14 + 5t, -3t, 1 + 7t\rangle$ and the line given by $\vec{r}(t) = \langle 3 - 3t, 5 + 2t, -2 + 4t\rangle$.
11. Does the line passing through $(-5, 4, -1)$ and $(-3, -5, 0)$ intersect the yz-plane? If so, give the point.
12. Does the line given by $\vec{r}(t) = \langle 6 + t, -8 + 14t, 4t\rangle$ intersect the xz-plane? If so, give the point.
13. Which of the three coordinate planes does the line given by $x = 16t$, $y = -4 - 9t$, $z = 34$ intersect?

12.3 Equations of Planes

For problems 1 - 5 write down the equation of the plane.

1. The plane containing the points $(6, -3, 1)$, $(5, -4, 1)$ and $(3, -4, 0)$.
2. The plane containing the point $(1, -5, 8)$ and orthogonal to the line given by $x = -3 + 15t$, $y = 14 - t$, $z = 9 - 3t$.
3. The plane containing the point $(-8, 3, 7)$ and parallel to the plane given by $4x + 8y - 2z = 45$.
4. The plane containing the point $(2, 0, -8)$ and containing the line given by $\vec{r}(t) = \langle 8t, -1 - 5t, 4 - t\rangle$.
5. The plane containing the two lines given by $\vec{r}(t) = \langle 7 + 5t, 2 + t, 6t\rangle$ and $\vec{r}(t) = \langle 7 - 6t, 2 - 2t, 10t\rangle$.

For problems 6 - 8 determine if the two planes are parallel, orthogonal or neither.

6. The plane given by $-5x + 3y + 2z = -8$ and the plane given by $6x - 8z = 15$.
7. The plane given by $3x + 9y + 7z = -1$ and the plane containing the points $(1, -1, 9)$, $(4, -1, 2)$ and $(-2, 3, 4)$.
8. The plane given by $-x - 8y + 3z = 6$ and the plane given by $2x + 2y + 6z = -91$.

For problems 9 - 11 determine where the line intersects the plane or show that it does not intersect the plane.

9. The line given by $\vec{r}(t) = \langle 9 + t, -4 + t, 2 + 5t\rangle$ and the plane given by $4x - 9y + z = 6$.
10. The line given by $\vec{r}(t) = \langle 2 - 3t, 1 + t, -4 - 2t\rangle$ and the plane given by $x - 7y - 4z = -1$.
11. The line given by $x = 8$, $y = -9t$, $z = 1 + 10t$ and the plane given by $8x + 9y + 2z = 17$.

For problems 12 & 13 find the line of intersection of the two planes.

12. Find the line of intersection of the plane given by $4x + y + 10z = -2$ and the plane given by $-8x + 2y + 3z = -8$.
13. Find the line of intersection of the plane given by $x - 10y - 2z = 3$ and the plane given by $2x - y + z = -13$.

14. Determine if the line given by $x = 4 + 3t$, $y = -2$, $z = 1 + 6t$ and the plane given by $8x - y + 4z = -3$ are parallel, orthogonal or neither.

12.4 Quadric Surfaces

Sketch each of the following quadric surfaces.

1. $z^2 = x^2 + \dfrac{y^2}{2}$
2. $z = 2 + 4x^2 + 6y^2$
3. $4x^2 + y^2 + 3z^2 = 1$
4. $\dfrac{x^2}{9} + \dfrac{y^2}{16} = 1$
5. $y = \dfrac{x^2}{2} + \dfrac{z^2}{3} - 7$
6. $6x^2 + 2z^2 = 1$
7. $x = 12 - \dfrac{y^2}{4} - 3z^2$
8. $x^2 = 4y^2 + 9z^2$

12.5 Functions of Several Variables

For problems 1 - 6 find the domain of the given function.

1. $f(x,y) = \sqrt{2x+4y-1}$
2. $f(x,y) = \ln\left(\dfrac{1}{x-y}\right)$
3. $f(x,y) = \sqrt{\dfrac{1}{x^2} - \dfrac{1}{y^2}}$
4. $f(x,y,z) = \dfrac{1}{x+1} + \dfrac{1}{y-1} + \dfrac{1}{x+y-z}$
5. $f(x,y,z) = \ln\left(x^2+y^2-8z\right)$
6. $f(x,y) = \sqrt{x+y} - \sqrt{x-3}$

For problems 7 - 11 identify and sketch the level curves (or contours) for the given function.

7. $x^2 - 4z - y = 2$
8. $x - 4z - y^2 = 2$
9. $z^2 + 4x^2 = 1 - 4y^2$
10. $z + 4x^2 = 1 - 4y^2$
11. $2x - 6y + z = -2$

For problems 12 - 14 identify and sketch the traces for the given curves.

12. $x^2 - 4z - y = 2$
13. $z^2 + 4x^2 = 1 - 4y^2$
14. $2x - 6y + z = -2$

12.6 Vector Functions

For problems 1 - 3 find the domain of the given vector function.

1. $\vec{r}(t) = \left\langle \frac{1}{t^2-1}, \frac{1}{t+3}, \frac{1}{t-6} \right\rangle$
2. $\vec{r}(t) = \left\langle \sqrt{t}, \sqrt{t+1}, \sqrt{t+2} \right\rangle$
3. $\vec{r}(t) = \langle \ln(t+7), \ln(t-3) \rangle$

For problems 4 - 8 sketch the graph of the given vector function.

4. $\vec{r}(t) = \langle -4, t+1 \rangle$
5. $\vec{r}(t) = \langle -2\cos(t), 5sin(t) \rangle$
6. $\vec{r}(t) = \left\langle \sqrt{t+2}, 1-t \right\rangle$
7. $\vec{r}(t) = \left\langle 2t+1, t^2-1 \right\rangle$
8. $\vec{r}(t) = \left\langle t^2+4, 6-t^2 \right\rangle$

For problems 9 - 12 identify the graph of the vector function without sketching the graph.

9. $\vec{r}(t) = \langle 6, 2+8t, -1+10t \rangle$
10. $\vec{r}(t) = \langle 12t, 6-8t, 4+7t \rangle$
11. $\vec{r}(t) = \langle 2, 6\cos(t), 6\sin(t) \rangle$
12. $\vec{r}(t) = \langle -2t, 6\cos(t), 6\sin(t) \rangle$

For problems 13 - 16 write down the equation of the line segment between the two points.

13. The line segment starting at $(4, -7)$ and ending at $(2, 0)$.
14. The line segment starting at $(-1, 2)$ and ending at $(7, -2)$.
15. The line segment starting at $(4, 1, -3)$ and ending at $(-1, 2, 6)$.
16. The line segment starting at $(1, -1, 9)$ and ending at $(4, -7, 10)$.

12.7 Calculus with Vector Functions

For problems 1 - 6 evaluate the given limit.

1. $\lim\limits_{t\to 0}\left(\cos(2t)\,\vec{i} - \mathbf{e}^{4-t}\vec{j} + \left(t^2+3t-9\right)\vec{k}\right)$

2. $\lim\limits_{t\to 4}\left\langle \dfrac{t-4}{t^2-3t-4}, \dfrac{t^2-4t}{16-t^2}\right\rangle$

3. $\lim\limits_{t\to 0}\left\langle \dfrac{\sin(t)}{2t}, \dfrac{1-\cos(t)}{t}, -3\right\rangle$

4. $\lim\limits_{t\to -8}\left(\dfrac{\mathbf{e}^{t^2-64}-1}{t+8}\,\vec{i} + \dfrac{\sin(t+8)}{t+8}\,\vec{j} - \vec{k}\right)$

5. $\lim\limits_{t\to -\infty}\left\langle \dfrac{5t^2-8t+1}{12+5t^2}, \dfrac{2+t^3}{1+t^2+t^4}\right\rangle$

6. $\lim\limits_{t\to \infty}\left\langle \ln\left(1-\dfrac{4}{t}\right), \mathbf{e}^{\frac{1}{t^2}}, 2\right\rangle$

For problems 7 - 11 compute the derivative of the given vector function.

7. $\vec{r}(t) = \left\langle \sqrt{3t}, \dfrac{1}{t^4}, \dfrac{1}{2t}\right\rangle$

8. $\vec{r}(t) = \cos(2t)\,\vec{i} - \sin(2t)\,\vec{j} + \ln(2t)\,\vec{k}$

9. $\vec{r}(t) = \left\langle \mathbf{e}^{t^2-1}, 4-\sec(2t), 7\right\rangle$

10. $\vec{r}(t) = \sin(t)\cos(t)\;\vec{i} - t^3\ln\left(t^2\right)\vec{j}$

11. $\vec{r}(t) = \left\langle \dfrac{1}{(t^2-4)^2}, \dfrac{t^3}{t^2+2}, \dfrac{t^2+2}{t^3}\right\rangle$

For problems 12 - 15 evaluate the given integral.

12. $\displaystyle\int \vec{r}(t)\;dt$, where $\vec{r}(t) = \left\langle \dfrac{1}{t^3} - t^3, 5t, \dfrac{1}{6t} - \dfrac{8}{t^4}\right\rangle$

13. $\displaystyle\int \vec{r}(t)\;dt$, where $\vec{r}(t) = \left(t^2-5\right)\mathbf{e}^{t^3-15t}\,\vec{i} + 4t\sqrt{t^2+1}\,\vec{j} - \sin^2(5t)\,\vec{k}$

14. $\displaystyle\int_0^1 \vec{r}(t)\;dt$ where $\vec{r}(t) = \left\langle t\cos(\pi t), 8t-2, 12t^3 - \mathbf{e}^{2t}\right\rangle$

15. $\displaystyle\int_{1}^{-1} \vec{r}(t)\ dt$ where $\vec{r}(t) = \tan(t)\ \vec{i} - \sin^3(t)\cos^2(t)\,\vec{j} - 8t$

12.8 Tangent, Normal and Binormal Vectors

For problems 1 - 3 find the unit tangent vector for the given vector function.

1. $\vec{r}(t) = t^2\,\vec{i} - \cos(8t)\,\vec{j} + \sin(8t)\,\vec{k}$
2. $\vec{r}(t) = \left\langle 8t, 2 - t^6, t^4 \right\rangle$
3. $\vec{r}(t) = \left\langle \ln(6t), \mathbf{e}^{1-t}, 5t \right\rangle$

For problems 4 & 5 find the tangent line to the vector function at the given point.

4. $\vec{r}(t) = \left\langle 3 + t^2, t^4, 6 \right\rangle$ at $t = -1$.
5. $\vec{r}(t) = \left\langle 2t, \cos^2(t), \mathbf{e}^{6t} \right\rangle$ at $t = 0$.

For problems 6 & 7 find the unit normal and the binormal vectors for the given vector function.

6. $\vec{r}(t) = \left\langle \mathbf{e}^{4t}\sin(t), \mathbf{e}^{4t}\cos(t), 2 \right\rangle$
7. $\vec{r}(t) = 2t\,\vec{i} + \frac{1}{2}t^2\,\vec{j} + \ln\left(t^2\right)\vec{k}$

12.9 Arc Length with Vector Functions

For problems 1 - 3 determine the length of the vector function on the given interval.

1. $\vec{r}(t) = 4\cos(2t)\,\vec{i} + 3t\,\vec{j} - 4\sin(2t)\,\vec{k}$ from $0 \le t \le 3\pi$.
2. $\vec{r}(t) = \left\langle 9 - 2t, 4 + 2t, \sqrt{2}\,t^2 \right\rangle$ from $0 \le t \le 1$.
3. $\vec{r}(t) = 2t\,\vec{i} + \frac{1}{2}t^2\,\vec{j} + \ln\left(t^2\right)\vec{k}$ from $1 \le t \le 3$.

For problems 4 - 6 find the arc length function for the given vector function.

4. $\vec{r}(t) = \langle 8t, 6 + t, -7t \rangle$
5. $\vec{r}(t) = \left\langle 8t, 4t^{\frac{3}{2}}, 3 \right\rangle$
6. $\vec{r}(t) = \left\langle \mathbf{e}^{4t}\sin(t), \mathbf{e}^{4t}\cos(t), 2 \right\rangle$
7. Determine where on the curve given by $\vec{r}(t) = \left\langle 8t, 4t^{\frac{3}{2}}, 3 \right\rangle$ we are after traveling a distance of 4.
8. Determine where on the curve given by $\vec{r}(t) = \left\langle \mathbf{e}^{4t}\sin(t), \mathbf{e}^{4t}\cos(t), 2 \right\rangle$ we are after traveling a distance of 15.

12.10 Curvature

Find the curvature for each the following vector functions.

1. $\vec{r}(t) = \left\langle 5t, 1-2t, 4t^{\frac{3}{2}} \right\rangle$
2. $\vec{r}(t) = \left\langle 6, \mathbf{e}^{-5t}, 3t\mathbf{e}^{-5t} \right\rangle$
3. $\vec{r}(t) = \langle \cos(\omega t), t, \sin(\omega t) \rangle$

12.11 Velocity and Acceleration

1. An objects acceleration is given by $\vec{a} = \cos(2t)\,\vec{i} + 4t^3\,\vec{j} + 6\sin(3t)\,\vec{k}$. The objects initial velocity is $\vec{v}(0) = 6\,\vec{i} + 2\vec{j} + 7\vec{k}$ and the objects initial position is $\vec{r}(0) = \vec{i} - 9\vec{j} + 6\vec{k}$. Determine the objects velocity and position functions.

2. An objects acceleration is given by $\vec{a} = 10t\,\vec{i} - 6\,\vec{j} + t\cos(t)\,\vec{k}$. The objects initial velocity is $\vec{v}(0) = -\vec{i} + 11\vec{j} - \vec{k}$ and the objects initial position is $\vec{r}(0) = 4\vec{i} + \vec{j} + 10\vec{k}$. Determine the objects velocity and position functions.

3. Determine the tangential and normal components of acceleration for the object whose position is given by $\vec{r}(t) = \left\langle 5t, 1 - 2t, 4t^{\frac{3}{2}} \right\rangle$.

4. Determine the tangential and normal components of acceleration for the object whose position is given by $\vec{r}(t) = \left\langle 6, \mathbf{e}^{-5t}, 3t\mathbf{e}^{-5t} \right\rangle$.

12.12 Cylindrical Coordinates

For problems 1 & 2 convert the Cartesian coordinates for the point into Cylindrical coordinates.

1. $(-3, 5, -8)$
2. $(4, 1, 7)$
3. Convert the following equation written in Cartesian coordinates into an equation in Cylindrical coordinates. $\dfrac{x-y}{x^2+y^2+1} = xyz$

For problems 4 - 6 convert the equation written in Cylindrical coordinates into an equation in Cartesian coordinates.

4. $zr^3 \cos(\theta) = 4r + 8$
5. $r^2 - 3\sin(\theta) = z^3 + \sqrt{r^2+1}$
6. $\tan(\theta) + 2z = 1 - r^2$

For problems 7 - 9 identify the surface generated by the given equation.

7. $z = -4r, \;\; z < 0$
8. $2r + 6\cos(\theta) + 18\sin(\theta) = \dfrac{51}{r}$
9. $\theta = \dfrac{\pi}{3}$

12.13 Spherical Coordinates

For problems 1 - 3 convert the Cartesian coordinates for the point into Spherical coordinates.

1. $(6, 2, -8)$
2. $(-1, 5, 2)$
3. $(-3, -2, 1)$
4. Convert the Cylindrical coordinates for the point $(5, 1.294, 6)$ into Spherical coordinates.
5. Convert the following equation written in Cartesian coordinates into an equation in Spherical coordinates.
$$\frac{xz}{y} = 2 - x$$

For problems 6 - 8 convert the equation written in Spherical coordinates into an equation in Cartesian coordinates.

6. $\rho \cos(\varphi) \sin(\varphi) \sin(\theta) = 3$
7. $\rho - \cos(\varphi) = 2 + \cos^2(\varphi)$
8. $\tan(\varphi) (\cos(\theta) - \sin(\theta)) = 4$

For problems 9 & 10 identify the surface generated by the given equation.

9. $\cos^2(\varphi) - \sin^2(\varphi) = 0$
10. $\sin(\varphi) cos\theta + sin\varphi sin\theta + \cos(\varphi) = \frac{1}{\rho}$

13 Partial Derivatives

To this point, with the exception of the occasional section in the last chapter, we've been working almost exclusively with functions of a single variable. It is not time to formally start multi-variable Calculus, *i.e.* Calculus involving functions of two or more variables. We will be covering the same basic topics as we do with single variable Calculus. Namely, limits, derivatives and integrals.

In this chapter we will open up with a quick section discussing taking limits of multi-variable functions. We will only be covering limits of multi-variable functions with a single chapter because, as we'll see, many of the concepts from single variable limits still hold, with some natural extensions of course. However, as we'll also see the work will often be significantly longer/harder and so we won't be spending a lot of time discussing limits of multi-variable functions. Luckily enough for us we also won't need to worry all that much about limits of multi-variable functions so the quick discussion of limits in this chapter will suffice.

The rest of the chapter will be discussing how to take derivatives of multi-variable functions. We want to keep the "main" interpretation of derivatives, namely the derivative will still give the rate of change of the function. The issue here is that because we have multiple variables the function can have differing rates of change depending on how we allow the various variables to change.

So, to start out the derivative discussion we will start by defining the partial derivative. These will restrict just how we allow the various variables to change. We will eventually introduce the directional derivative which will allow the variables to change in any arbitrary manner. In the process of introducing the idea of a directional derivative we'll also introduce the concept of a gradient of a function. The gradient will arise in quite a few sections throughout the rest of this multi-variable Calculus material, including integrals.

Finally, as we'll see, if you can take derivatives of single variable functions then you have the majority of the knowledge that you need to take derivatives of multi-variable functions. There are, however, some subtleties that we'll need to remember to deal with. Those subtleties are, generally, the issues that most students run into when taking derivatives of multi-variable functions.

If you are looking for some practice problems (with solutions available) please check out the Practice Problems. There you will find a set of problems that should give you quite a bit practice.

13.1 Limits

Evaluate each of the following limits.

1. $\displaystyle\lim_{(x,y,z)\to(2,1,0)} \frac{\left(4y - z^3\right)\mathbf{e}^{3x-6}}{4z - yx^2}$

2. $\displaystyle\lim_{(x,y)\to(3,-7)} \frac{6x - y + xy}{2x^3 + y^3}$

3. $\displaystyle\lim_{(x,y)\to(-3,4)} \frac{4x^2 - xy - 3y^2}{12x^2 + 17xy + 6y^2}$

4. $\displaystyle\lim_{(x,y)\to(-1,10)} \frac{10x^2 + 11xy + y^2}{10x^2 - 39xy - 4y^2}$

5. $\displaystyle\lim_{(x,y)\to(0,0)} \frac{2x^2 + 7y^2}{4y^2 + x^2}$

6. $\displaystyle\lim_{(x,y)\to(0,0)} \frac{6\sqrt[3]{x} - 3y^{10}}{9y^{30} + 2x}$

7. $\displaystyle\lim_{(x,y)\to(0,0)} \frac{2x^4y}{x^8 + 6y^2}$

13.2 Partial Derivatives

For problems 1 - 13 find all the 1st order partial derivatives.

1. $f(x,y,z) = x^3\sqrt{y} + 4z^3y^2 - xyz + x^2 - \sqrt[3]{z^5}$
2. $W(a,b,c,d) = a^2 + b^3 - c^2d^4 - 5a^3c - d^6b^2a$
3. $A(p,t,u) = \dfrac{1}{p\,t^2} - \dfrac{t^3}{u^2} + \dfrac{4u\,p}{t^4}$
4. $g(x,y,z) = \sqrt{x^2 + z^{-2}} + \sin\left(xy - x^2\right)$
5. $f(s,t) = \cos\left(s\mathbf{e}^{t^2}\right) + \cos\left(s + \mathbf{e}^{t^2}\right)$
6. $f(x,y) = \ln\left(\dfrac{y}{x}\right) + \ln\left(\dfrac{1}{x+y}\right) - \ln\left(\dfrac{x}{6}\right)$
7. $A(y,z) = \dfrac{1}{y - 4z^5} + \tan\left(yz^2 - y^3\right)$
8. $g(u,v) = \dfrac{u}{v}\cos\left(\dfrac{v}{u}\right) + 4u - v^2u$
9. $w = (x-y)\,\mathbf{e}^{4x+z^6} - \sin(2x+7z)\sec\left(yz^3\right)$
10. $f(u,v,w) = (uw+4)\sin^{-1}\left(u^2+v^2\right) - \ln\left(\frac{w^2}{v^4}\right)$
11. $f(x,y,z) = \sin\left(\dfrac{z}{z^2+x}\right) - \dfrac{6x^2+y}{y^2-z^2}$
12. $g(s,t,p) = \dfrac{p^3t^2}{s^2+1} + \dfrac{(4s-1)\,t^2}{6-s}$
13. $f(x,y,z,w) = x^2\sin(4y) + z^3(6x-y) + y^4$

For problems 14 & 15 find $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$ for the given function.

14. $z^4 - y^2 + x^2 = 6x^2y^3z^7$
15. $x^2\sin(z) + \left(x^2-1\right)y^4 = z^6$

13.3 Interpretations of Partial Derivatives

1. Determine if $f(x,y) = 10 - x^2 - y^2$ is increasing or decreasing at $(7,-3)$ if

   (a) we allow x to vary and hold y fixed.

   (b) we allow y to vary and hold x fixed.

2. Determine if $f(x,y) = x\mathbf{e}^{x-y} + 100y$ is increasing or decreasing at $(-2,1)$ if

   (a) we allow x to vary and hold y fixed.

   (b) we allow y to vary and hold x fixed.

3. Determine if $f(x,y) = \dfrac{x+y}{y-x}$ is increasing or decreasing at $(0,7)$ if

   (a) we allow x to vary and hold y fixed.

   (b) we allow y to vary and hold x fixed.

4. Write down the vector equations of the tangent lines to the traces for $f(x,y) = \sin(x)\cos(y)$ at $\left(\dfrac{\pi}{3}, -\dfrac{\pi}{4}\right)$.

5. Write down the vector equations of the tangent lines to the traces for $f(x,y) = \ln\left(\dfrac{x}{y^2}\right)$ at $(6,2)$.

13.4 Higher Order Partial Derivatives

For problems 1 - 3 verify Clairaut's Theorem for the given function.

1. $Q(s,t) = \ln(st) - s^4 \sin(6t) + st$
2. $f(u,w) = \sin(uw) + 4u^2 w^{-2}$
3. $f(x,y) = \mathbf{e}^{xy} \sin(y)$

For problems 4 - 9 find all 2nd order derivatives for the given function.

4. $h(x,y) = x^4 y^{-2} - 4xy + \mathbf{e}^{7y} + \ln(2x)$
5. $A(u,v) = u^2 \cos(3v) + \ln\left(\frac{u}{4v^2}\right)$
6. $g(v,w) = \ln(v \sin(w)) + \sin(v \ln(w))$
7. $f(x,y) = \cos\left(x^2 + y^2\right) - \sin(xy)$
8. $h(x,y,z) = 7x^3 y^2 z^4 + 8$
9. $Q(u,v,w) = u^4 \sin\left(w^2\right) - \frac{2v}{u^4} + \ln\left(v^2 w\right)$

For problems 10 & 11 find all 3rd order derivatives for the given function.

10. $h(x,y) = x^4 y^5 - 5\sqrt{x} + 8y^2$
11. $A(u,v) = u^3 \sin(2v) - \frac{u^3}{v^2}$
12. Given $f(x,y,z) = \mathbf{e}^{-z} \cos(4y) \ln(2x)$ find f_{xyyzxz}.
13. Given $w = \ln\left(\frac{xy}{z}\right) + 8x^4 y^3 \sqrt{z}$ find $\frac{\partial^5 w}{\partial x \partial z^2 \partial y \partial x}$.
14. Given $h(u,v) = \cos\left(u^4 + u^2 + 1\right) - \frac{\sqrt{u}}{v^3}$ find $\frac{\partial^7 h}{\partial u^2 \partial v \partial u^4}$.
15. Given $f(x,y) = \frac{x^6}{1+6y} - \cos\left(x^2\right) + 6\mathbf{e}^x \sin(y)$ find f_{xxyxyx}.

13.5 Differentials

Compute the differential of each of the following functions.

1. $w = \dfrac{x^4 z^8}{y}$
2. $f(x, y) = \tan\left(xy^2\right)$
3. $A(x, y, z) = z^3 \mathbf{e}^{x\,z} \sec(y)$

13.6 Chain Rule

1. Given the following information use the Chain Rule to determine $\dfrac{dz}{dt}$.

$$z = \mathbf{e}^{x^2 - y} \qquad x = \sin(4t), \quad y = t^3 - 9$$

2. Given the following information use the Chain Rule to determine $\dfrac{dw}{dt}$.

$$w = x^4 - 4xy^2 + z^3 \qquad x = \sqrt{t}, \quad y = \mathbf{e}^{2t}, \quad z = \frac{1}{t}$$

3. Given the following information use the Chain Rule to determine $\dfrac{dw}{dt}$.

$$w = \frac{4x}{y\,z^3} \qquad x = 7t - 1, \quad y = 1 - 2t, \quad z = t^4$$

4. Given the following information use the Chain Rule to determine $\dfrac{dz}{dx}$.

$$z = 2x^3\mathbf{e}^{4y} \qquad y = \cos(6x)$$

5. Given the following information use the Chain Rule to determine $\dfrac{dz}{dx}$.

$$z = \tan\left(\frac{x}{y}\right) \qquad y = \mathbf{e}^{x^2}$$

6. Given the following information use the Chain Rule to determine $\dfrac{\partial z}{\partial u}$ and $\dfrac{\partial z}{\partial v}$.

$$z = x\sin\left(y^2 - x\right) \qquad x = 3u - v^2, \quad y = u^6$$

7. Given the following information use the Chain Rule to determine w_u and w_v .

$$w = x^4y^{-3}z^2 \qquad x = u^2v, \quad y = 3 - uv, \quad z = 7u^2 - 10v$$

8. Given the following information use the Chain Rule to determine $\dfrac{\partial z}{\partial t}$ and $\dfrac{\partial z}{\partial s}$.

$$z = 6x + y^2\tan(x) \qquad x = p^2 - 3t, \quad y = s^2 - t^2, \quad p = \mathbf{e}^{3s}$$

9. Given the following information use the Chain Rule to determine w_p and w_t .

$$w = x^2y^4z^6 - 2xy \qquad x = 2p, \quad y = 3tq, \quad z = 3tp^2, \quad q = 2t$$

10. Given the following information use the Chain Rule to determine $\dfrac{\partial w}{\partial u}$ and $\dfrac{\partial w}{\partial v}$.

$$w = \frac{\sqrt{y}}{x^2 z^3} \qquad x = uv, \quad y = u^2 - p^3, \quad z = 4qp, \quad p = 2u - 3v, \quad q = v^2$$

11. Determine formulas for w_u and w_t for the following situation.

$$w = w\left(x, y\right) \qquad x = x\left(y, z\right), \quad y = y\left(u, v\right), \quad z = z\left(u, t\right), \quad v = v\left(t\right)$$

12. Determine formulas for $\dfrac{\partial w}{\partial s}$ and $\dfrac{\partial w}{\partial t}$ for the following situation.

$$w = w\left(x, y, z\right) \qquad x = x\left(u, v, t\right), \quad y = y\left(p\right), \quad z = z\left(u, t\right), \quad v = v\left(p, t\right), \quad p = p\left(s, t\right)$$

13. Compute $\dfrac{dy}{dx}$ for the following equation.

$$\cos\left(2x + 3y\right) = x^5 - 8y^2$$

14. Compute $\dfrac{dy}{dx}$ for the following equation.

$$\cos\left(2x\right)\sin\left(3y\right) - xy = y^4 + 9$$

15. Compute $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$ for the following equation.

$$z^3 y^4 - x^2 \cos\left(2y - 4z\right) = 4z$$

16. Compute $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$ for the following equation.

$$\sin\left(x\right)\mathbf{e}^{4x\,z} + 2z^2 y = \cos\left(z\right)$$

17. Determine $f_{u\,u}$ and $f_{v\,v}$ for the following situation.

$$f = f\left(x, y\right) \qquad x = \mathbf{e}^u sin\left(v\right), \quad y = \mathbf{e}^u \cos\left(v\right)$$

18. Determine $f_{u\,u}$ and $f_{v\,v}$ for the following situation.

$$f = f\left(x, y\right) \qquad x = u^2 - v^2, \quad y = \frac{u}{v}$$

13.7 Directional Derivatives

For problems 1 - 4 determine the gradient of the given function.

1. $f(x,y) = y^3x^5 + \ln(xy)$
2. $f(x,y) = \mathbf{e}^{\frac{x}{y}} + y^4 \sin(xy)$
3. $f(x,y,z) = 4z - \dfrac{y^4}{2z^3} + \sqrt{x^3}(z-1)$
4. $f(x,y,z) = \cos\left(\dfrac{xy}{z}\right) + z^3y^2x$

For problems 5 - 8 determine $D_{\vec{u}}f$ for the given function in the indicated direction.

5. $f(x,y) = \ln(2xy) - \sin\left(x^2+y^2\right)$ in the direction of $\vec{v} = \langle 7,-3\rangle$
6. $f(x,y) = 4x^2y^3 - \sqrt{2x+5y}$ in the direction of $\vec{v} = \langle -1,4\rangle$
7. $f(x,y,z) = 8xy^2 - \dfrac{5z^2}{x} + y^4$ in the direction of $\vec{v} = \langle -4,1,2\rangle$
8. $f(x,y,z) = \dfrac{3x}{y^2-z^3} + 5x^2 - 8y$ in the direction of $\vec{v} = \langle 0,3,-2\rangle$
9. Determine $D_{\vec{u}}f(-1,4,6)$ for $f(x,y,z) = \mathbf{e}^{x\,y^2} + 4zy^3$ direction of $\vec{v} = \langle 2,-3,6\rangle$.
10. Determine $D_{\vec{u}}f(8,1,2)$ for $f(x,y,z) = \ln\left(\dfrac{x}{z}\right) + \ln\left(\dfrac{z}{y}\right) + y^2x$ direction of $\vec{v} = \langle 1,5,2\rangle$.

For problems 11 - 13 find the maximum rate of change of the function at the indicated point and the direction in which this maximum rate of change occurs.

11. $f(x,y) = \mathbf{e}^{4x\,y}$ at $(6,-2)$
12. $f(x,y,z) = x^2y^4 - 3z^2x$ at $(1,-6,3)$
13. $f(x,y,z) = \ln\left(\dfrac{2x+3y}{z}\right)$ at $(2,7,4)$
14. Given $\vec{u} = \left\langle -\dfrac{3}{5}, -\dfrac{4}{5}\right\rangle$, $\vec{v} = \left\langle \dfrac{4}{\sqrt{20}}, \dfrac{2}{\sqrt{20}}\right\rangle$, $\vec{w} = \left\langle -\dfrac{3}{\sqrt{11}}, -\dfrac{2}{\sqrt{11}}\right\rangle$, $D_{\vec{u}}(-1,4) = \dfrac{14}{5}$ and $D_{\vec{v}}(-1,4) = -\dfrac{22}{\sqrt{20}}$ determine the value of $D_{\vec{w}}(-1,4)$.

15. Given $\vec{u} = \left\langle \frac{1}{\sqrt{15}}, \frac{4}{\sqrt{15}} \right\rangle$, $\vec{v} = \left\langle -\frac{3}{\sqrt{34}}, -\frac{5}{\sqrt{34}} \right\rangle$, $\vec{w} = \left\langle -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$, $D_{\vec{u}}(0,1) = \frac{18}{\sqrt{15}}$ and $D_{\vec{v}}(0,1) = -\frac{40}{\sqrt{34}}$ determine the value of $D_{\vec{w}}(0,1)$.

14 Applications of Partial Derivatives

In this chapter we'll take a look at a couple of applications of partial derivatives. The applications here are either very similar to applications we saw for derivatives of single variable functions or extensions of those applications.

For example we will be looking at the tangent plane to a surface rather than tangent lines to curves as we did with single variable functions.

In addition we be finding relative and absolute extrema of multi-variable functions. The difference in this chapter compared to the last time we saw these applications is that they will often involve a lot more work. Because of the increased difficulty of the problems we'll be restricting ourselves to finding the relative and absolute extrema of functions of two variables only.

We will also be looking at Lagrange Multipliers. This is a method that will allow us to optimize a function that is subject to some constraint. That is to say optimizing a function of two or three variables where the variables must also satisfy some constraint (usually in the form on an equation involving the variables).

If you are looking for some practice problems (with solutions available) please check out the Practice Problems. There you will find a set of problems that should give you quite a bit practice.

14.1 Tangent Planes and Linear Approximations

1. Find the equation of the tangent plane to $z = x^2y^4 - \dfrac{12x}{y}$ at $(-1, 6)$.

2. Find the equation of the tangent plane to $z = \ln\left(x^2y\right) - x\sqrt{y}$ at $\left(-\dfrac{1}{2}, 4\right)$.

3. Find the equation of the tangent plane to $z = \mathbf{e}^{x\,y} + y^2\mathbf{e}^{1-y}$ at $(0, 1)$.

4. Find the linear approximation to $z = \cos\left(\sin\left(y\right) - x\right)$ at $(-2, 0)$.

5. Find the linear approximation to $z = \dfrac{10x^2}{x - y}$ at $(4, -1)$.

14.2 Gradient Vector, Tangent Planes and Normal Lines

1. Find the tangent plane and normal line to $z\,y + 4x\sqrt{z} - x^3y^2 = 221$ at $(-2, 5, 9)$.

2. Find the tangent plane and normal line to $\mathbf{e}^{x\,y^2} + zy^4 = 61 + \dfrac{z^2}{x+1}$ at $(0, -2, 6)$.

3. Find the tangent plane and normal line to $9yz - \sqrt{x^2 - 8z} = xy^2 - 26$ at $(3, 1, -2)$.

4. Find the point(s) on $6x^2 + y^2 - 3z^2 = 4$ where the tangent plane to the surface is parallel to the plane given by $2x + 7y - z = 6$.

5. Find the point(s) on $x^2 - 8y^2 - 2z^2 = -3$ where the tangent plane to the surface is parallel to the plane given by $-4x - y + 8z = 1$.

14.3 Relative Minimums and Maximums

Find and classify all the critical points of the following functions.

1. $f(x,y) = 2y - 9x - xy + 5x^2 + y^2$
2. $f(x,y) = x^3 - y^3 + 8xy$
3. $f(x,y) = (y-x)(1-2x-3y)$
4. $f(x,y) = \frac{1}{2}x^4 - 4xy^2 - 2x^2 + 8y^2$
5. $f(x,y) = xy\,\mathbf{e}^{-8(x^2+y^2)}$
6. $f(x,y) = 8x - x\sqrt{y-1} + x^3 + \frac{1}{2}y - 12x^2$

14.4 Absolute Extrema

1. Find the absolute minimum and absolute maximum of $f(x,y) = 18x^2 + 4y^2 - y^2x - 2$ on the triangle with vertices $(-1,-1)$, $(5,-1)$ and $(5,17)$.

2. Find the absolute minimum and absolute maximum of $f(x,y) = 2x^3 - 4y^3 + 24xy$ on the rectangle given by $0 \le x \le 5$, $-3 \le y \le -1$.

3. Find the absolute minimum and absolute maximum of $f(x,y) = x^2 - y^2 + xy - 5x$ on the region bounded by $y = 5 - x^2$ and the x-axis.

14.5 Lagrange Multipliers

1. Find the maximum and minimum values of $f(x,y) = 10y^2 - 4x^2$ subject to the constraint $x^4 + y^4 = 1$.

2. Find the maximum and minimum values of $f(x,y) = 3x - 6y$ subject to the constraint $4x^2 + 2y^2 = 25$.

3. Find the maximum and minimum values of $f(x,y) = xy$ subject to the constraint $x^2 - y = 12$. Assume that $y \leq 0$ for this problem. Why is this assumption needed?

4. Find the maximum and minimum values of $f(x,y,z) = x^2 + 3y^2$ subject to the constraint $x^2 + 4y^2 + z^2 = 36$.

5. Find the maximum and minimum values of $f(x,y,z) = xyz$ subject to the constraint $x^2 + 2y^2 + 4z^2 = 24$.

6. Find the maximum and minimum values of $f(x,y,z) = 2x + 4y + z^2$ subject to the constraints $y^2 + z^2 = 1$ and $x^2 + z^2 = 1$.

7. Find the maximum and minimum values of $f(x,y,z) = x + y + z^2$ subject to the constraints $x + y + z = 1$ and $x^2 + z^2 = 1$.

15 Multiple Integrals

We now need to start discussing integration of multi-variable functions.

When we looked at definite integrals of single variable functions the values of the independent variable were in some interval $[a, b]$. For functions of multiple variables the values of the independent variables will not just come from intervals anymore. For functions of two variables, for example, the values of the independent variables will come from a two dimensional region. Likewise, for functions of three variables the values of the independent variables will come from a three dimensional region.

We will be discussing Double Integrals (for integrating functions of two variables) and Triple Integrals (for integrating functions of three variables). While most of the integration will be done in terms of Cartesian coordinates we will also discuss converting integrals from Cartesian coordinates into Polar coordinates (for functions of two variables) and Cylindrical or Spherical coordinates (for functions of three variables).

We will also formalize the process for converting an integral from one coordinate system into another. In the process we will derive some of the formulas that were using to convert integrals from Cartesian into Polar, Cylindrical or Spherical coordinates.

If you are looking for some practice problems (with solutions available) please check out the Practice Problems. There you will find a set of problems that should give you quite a bit practice.

15.1 Double Integrals

1. Use the Midpoint Rule to estimate the volume under $f(x, y) = 4x+8y$ and above the rectangle given by $0 \leq x \leq 4$, $2 \leq y \leq 6$ in the xy-plane. Use 4 subdivisions in the x direction and 4 subdivisions in the y direction.

2. Use the Midpoint Rule to estimate the volume under $f(x, y) = 4x-y^2$ and above the rectangle given by $0 \leq x \leq 2$, $-2 \leq y \leq 1$ in the xy-plane. Use 2 subdivisions in the x direction and 3 subdivisions in the y direction.

15.2 Iterated Integrals

1. Compute the following double integral over the indicated rectangle **(a)** by integrating with respect to x first and **(b)** by integrating with respect to y first.

$$\iint\limits_R 16xy - 9x^2 + 1\,dA \qquad R = [2,3] \times [-1,1]$$

2. Compute the following double integral over the indicated rectangle **(a)** by integrating with respect to x first and **(b)** by integrating with respect to y first.

$$\iint\limits_R \cos(x)\sin(y)\;dA \qquad R = \left[\frac{\pi}{6}, \frac{\pi}{4}\right] \times \left[\frac{\pi}{4}, \frac{\pi}{3}\right]$$

For problems 3 - 16 compute the given double integral over the indicated rectangle.

3. $\displaystyle\iint\limits_R 8x^3 - 4\,dA \qquad R = [-3,-1] \times [0,4]$

4. $\displaystyle\iint\limits_R 15y^4 + \frac{2}{x^2}\,dA \qquad R = [1,2] \times [1,4]$

5. $\displaystyle\iint\limits_R 4y\sec^2(x) + \frac{2x}{y}\,dA \qquad R = \left[0, \frac{\pi}{4}\right] \times [1,5]$

6. $\displaystyle\iint\limits_R y^5 - x^2\mathbf{e}^y\,dA \qquad R = [-1,2] \times [-3,3]$

7. $\displaystyle\iint\limits_R \frac{x^3}{1+x^4} - \frac{1}{\mathbf{e}^{3y}}\,dA \qquad R = [-1,0] \times [0,4]$

8. $\displaystyle\iint\limits_R x\mathbf{e}^{x^2} - 12x^3\sin(\pi y)\;dA \qquad R = [-2,0] \times \left[\frac{1}{2}, 1\right]$

9. $\displaystyle\iint\limits_R x\cos\left(4y + 3x^2\right)\;dA \qquad R = \left[0, \sqrt{\pi}\right] \times \left[\frac{\pi}{2}, \pi\right]$

10. $\displaystyle\iint\limits_R \frac{\ln(4xy)}{xy}\,dA \qquad R = [1,2] \times [3,4]$

11. $\displaystyle\iint\limits_R x^2y^3\,\mathbf{e}^{x^3 - y^4}\,dA \qquad R = [0,1] \times [-1,0]$

12. $\displaystyle\iint_R 42yx^3\left(1+x^2y^2\right)^6\,dA \qquad R=[0,1]\times[0,2]$

13. $\displaystyle\iint_R \frac{\cos\left(\frac{x}{y}\right)}{y^3}\,dA \qquad R=\left[\frac{\pi}{2},\pi\right]\times[1,2]$

14. $\displaystyle\iint_R \frac{x\cos\left(\frac{x}{y}\right)}{y^2}\,dA \qquad R=\left[\frac{\pi}{2},\pi\right]\times[1,2]$

15. $\displaystyle\iint_R 2y\ln(x)-20x^3y^3\,dA \qquad R=[1,2]\times[0,4]$

16. $\displaystyle\iint_R xy\mathbf{e}^x\cos(y)\,dA \qquad R=[0,1]\times[0,\pi]$

17. Determine the volume that lies under $f(x,y)=20-3x^3-3y^2$ and above the rectangle given by $[-2,2]\times[-1,1]$ in the xy-plane.

18. Determine the volume that lies under $f(x,y)=10+xy\sin\left(x^2-y^2\right)$ and above the rectangle given by $[-3,0]\times[1,3]$ in the xy-plane.

15.3 Double Integrals over General Regions

1. Evaluate $\iint\limits_D 8yx^3\,dA$ where $D = \left\{(x,y)\,|\,-1 \le y \le 2, -1 \le x \le 1+y^2\right\}$

2. Evaluate $\iint\limits_D 12x^2y - y^2\,dA$ where $D = \left\{(x,y)\,|\,-2 \le x \le 2, -x^2 \le y \le x^2\right\}$

3. Evaluate $\iint\limits_D 9 - \dfrac{6y^2}{x^2}\,dA$ where D is the region in the 1st quadrant bounded by $y = x^3$ and $y = 4x$.

4. Evaluate $\iint\limits_D 15x^2 - 6y\,dA$ where D is the region bounded by $x = \dfrac{1}{2}y^2$ and $x = 4\sqrt{y}$.

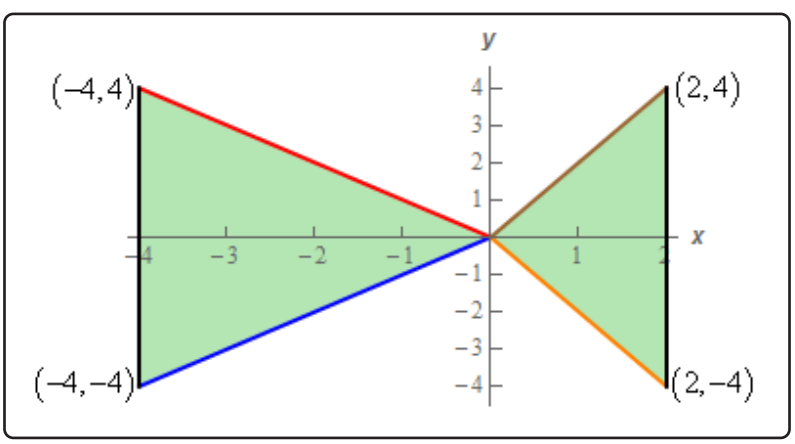
5. Evaluate $\iint\limits_D 6y(x+6)^2\,dA$ where D is the region bounded by $x = -y^2$ and $x = y - 6$.

6. Evaluate $\iint\limits_D \mathbf{e}^{y^2+1}\,dA$ where D is the triangle with vertices $(0,0)$, $(-2,4)$ and $(8,4)$.

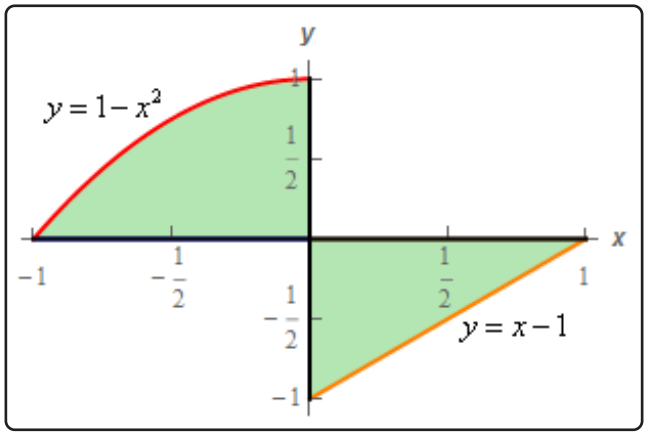
7. Evaluate $\iint\limits_D 7y^3\mathbf{e}^{x^2+1}\,dA$ where D is the region bounded by $y = 2\sqrt[4]{x}$, $x = 9$ and the x-axis.

8. Evaluate $\iint\limits_D x^5 \sin\left(y^4\right)\,dA$ where D is the region in the 2nd quadrant bounded by $y = 3x^2$, $y = 12$ and the y-axis.

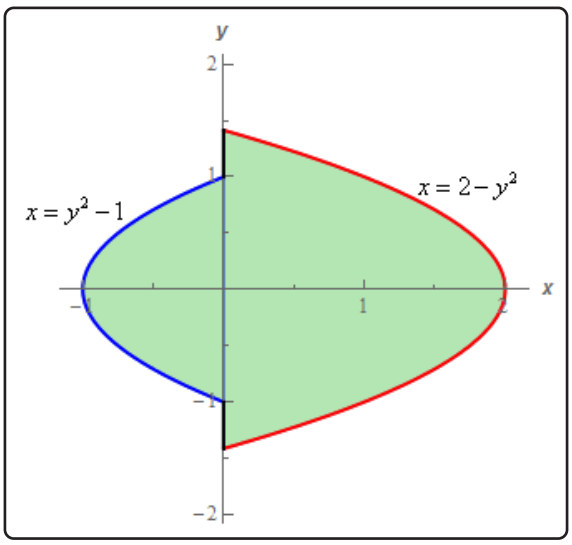
9. Evaluate $\iint\limits_D xy - y^2\,dA$ where D is the region shown below.



10. Evaluate $\iint\limits_D 12x^3 - 3\,dA$ where D is the region shown below.



11. Evaluate $\iint\limits_D 6y^2 + 10yx^4\,dA$ where D is the region shown below.



12. Evaluate $\iint\limits_D \frac{x^3}{y^2}\,dA$ where D is the region bounded by $y = \frac{1}{x^2}$, $x = 1$ and $y = \frac{1}{4}$ in the order given below.

(a) Integrate with respect to x first and then y.

(b) Integrate with respect to y first and then x.

13. Evaluate $\iint\limits_D xy - y^3 \, dA$ where D is the region bounded by $y = x^2$, $y = -x^2$ and $x = 2$ in the order given below.

    (a) Integrate with respect to x first and then y.

    (b) Integrate with respect to y first and then x.

For problems 14 - 16 evaluate the given integral by first reversing the order of integration.

14. $\displaystyle\int_0^8 \int_{y^{\frac{1}{3}}}^2 \frac{y}{x^7 + 1} \, dx \, dy$

15. $\displaystyle\int_{-4}^0 \int_{\sqrt{-x}}^2 x^{-\frac{2}{3}} \sqrt{y^{\frac{5}{3}} + 1} \, dy \, dx$

16. $\displaystyle\int_0^2 \int_{-x}^{3x} 5y^2x^3 + 2 \, dy \, dx$

17. Use a double integral to determine the area of the region bounded by $x = -y^2$ and $x = y - 6$.

18. Use a double integral to determine the area of the region bounded by $y = x^2 + 1$ and $y = \dfrac{1}{2}x^2 + 3$.

19. Use a double integral to determine the volume of the region that is between the xy-plane and $f(x, y) = 2 - xy^2$ and is above the region in the xy-plane that is bounded by $y = x^2$ and $x = 1$.

20. Use a double integral to determine the volume of the region that is between the xy-plane and $f(x, y) = 1 + y^5 \sqrt{x^4 + 1}$ and is above the region in the xy-plane that is bounded by $y = \sqrt{x}$, $x = 2$ and the x-axis.

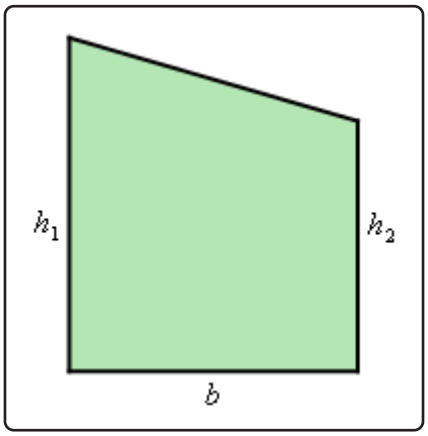
21. Use a double integral to determine the volume of the region in the first octant that is below the plane given by $2x + 6y + 4z = 8$.

22. Use a double integral to determine the volume of the region bounded by $z = 3 - 2y$, the surface $y = 1 - x^2$ and the planes $y = 0$ and $z = 0$.

23. Use a double integral to determine the volume of the region bounded by the planes $z = 4 - 2x - 2y$, $y = 2x$, $y = 0$ and $z = 0$.

24. Use a double integral to determine the formula for the area of a right triangle with base, b and height h.

25. Use a double integral to determine a formula for the figure below.



15.4 Double Integrals in Polar Coordinates

1. Evaluate $\iint\limits_D 3xy^2 - 2\,dA$ where D is the unit circle centered at the origin.

2. Evaluate $\iint\limits_D 4x - 2y\,dA$ where D is the top half of region between $x^2 + y^2 = 4$ and $x^2 + y^2 = 25$.

3. Evaluate $\iint\limits_D 6xy + 4x^2\,dA$ where D is the portion of $x^2 + y^2 = 9$ in the 2nd quadrant.

4. Evaluate $\iint\limits_D \sin\left(3x^2 + 3y^2\right)\,dA$ where D is the region between $x^2 + y^2 = 1$ and $x^2 + y^2 = 7$.

5. Evaluate $\iint\limits_D \mathbf{e}^{1-x^2-y^2}\,dA$ where D is the region in the 4th quadrant between $x^2 + y^2 = 16$ and $x^2 + y^2 = 36$.

6. Use a double integral to determine the area of the region that is inside $r = 6 - 4\cos(\theta)$.

7. Use a double integral to determine the area of the region that is inside $r = 4$ and outside $r = 8 + 6\sin(\theta)$.

8. Evaluate the following integral by first converting to an integral in polar coordinates.

$$\int_{-2}^{0}\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} x^2\,dx\,dy$$

9. Evaluate the following integral by first converting to an integral in polar coordinates.

$$\int_{-1}^{1}\int_{0}^{\sqrt{1-x^2}} \sqrt{x^2 + y^2}\,dy\,dx$$

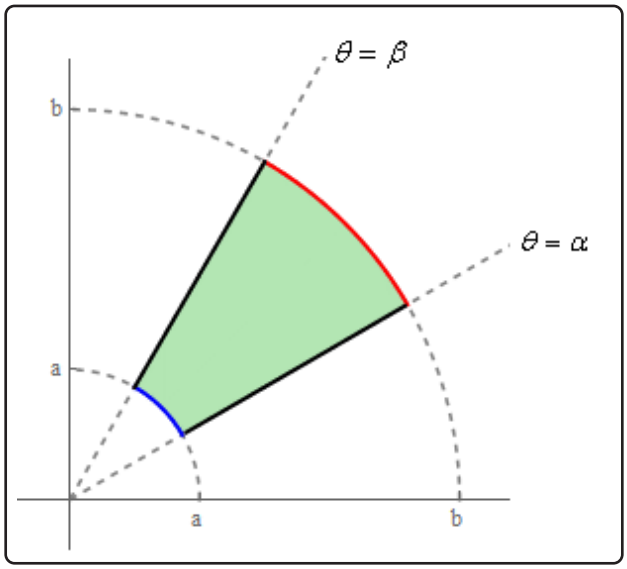
10. Use a double integral to determine the volume of the solid that is below $z = 9 - 4x^2 - 4y^2$ and above the xy-plane.

11. Use a double integral to determine the volume of the solid that is bounded by $z = 12 - 3x^2 - 3y^2$ and $z = x^2 + y^2 - 8$.

12. Use a double integral to determine the volume of the solid that is inside both the cylinder $x^2 + y^2 = 9$ and the sphere $x^2 + y^2 + z^2 = 16$.

13. Use a double integral to derive the area of a circle of radius a.

14. Use a double integral to derive the area of the region between circles of radius a and b with $\alpha \leq \theta \leq \beta$. See the image below for a sketch of the region.



15.5 Triple Integrals

1. Evaluate $\int_1^2 \int_0^2 \int_{-1}^1 2 + z^2 - xy \, dz \, dx \, dy$

2. Evaluate $\int_2^0 \int_{x^2}^1 \int_0^{xz} y^2 - 6z \, dy \, dz \, dx$

3. Evaluate $\int_{-1}^2 \int_0^1 \int_0^{2z} 3x - \sqrt{1+z^2} \, dx \, dz \, dy$

4. Evaluate $\iiint\limits_E 12y \, dV$ where E is the region below $6x + 4y + 3z = 12$ in the first octant.

5. Evaluate $\iiint\limits_E 5x^2 \, dV$ where E is the region below $x + 2y + 4z = 8$ in the first octant.

6. Evaluate $\iiint\limits_E 10z^2 - x \, dV$ where E is the region below $z = 8 - y$ and above the region in the xy-plane bounded by $y = 2x$, $x = 3$ and $y = 0$.

7. Evaluate $\iiint\limits_E 4y^2 \, dV$ where E is the region below $z = -3x^2 - 3y^2$ and above $z = -12$.

8. Evaluate $\iiint\limits_E 2y - 9z \, dV$ where E is the region behind $6x + 3y + 3z = 15$ front of the triangle in the xy-plane with vertices, in (x, z) form :$(0, 0)$, $(0, 4)$ and $(2, 4)$.

9. Evaluate $\iiint\limits_E 18x \, dV$ where E is the region behind the surface $y = 4 - x^2$ that is in front of the region in the xz-plane bounded by $z = -3x$, $z = 2x$ and $z = 2$.

10. Evaluate $\iiint\limits_E 20x^3 \, dV$ where E is the region bounded by $x = 2 - y^2 - z^2$ and $x = 5y^2 + 5z^2 - 6$.

11. Evaluate $\iiint\limits_E 6z^2 \, dV$ where E is the region behind $x + 6y + 2z = 8$ that is in front of the region in the xy-plane bounded by $z = 2y$ and $z = \sqrt{4y}$.

12. Evaluate $\iiint\limits_E 8y \, dV$ where E is the region between $x + y + z = 6$ and $x + y + z = 10$ above the triangle in the xy-plane with vertices, in (x, y) form : $(0, 0)$, $(1, 2)$ and $(1, 4)$.